

An Introduction to the Smith Ecophysiology Lab: Physiology for Understanding Larger Scale Processes

Nick Smith

School of Biological Sciences

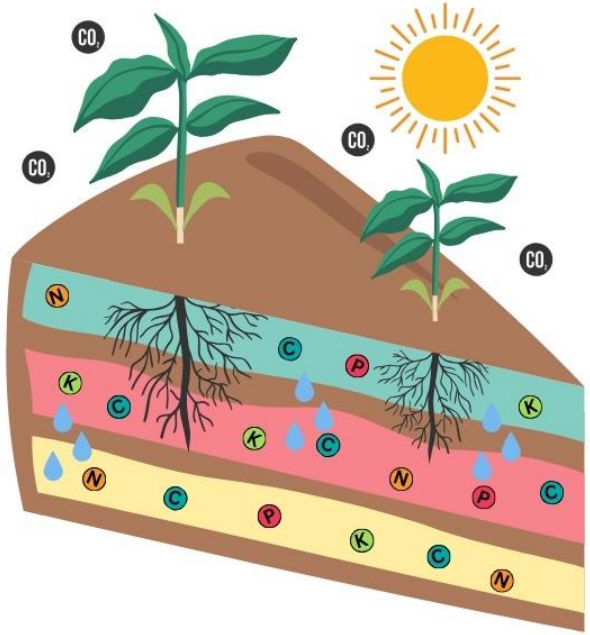
University of Nebraska - Lincoln

nsmith87@unl.edu

PhUnKY Lab

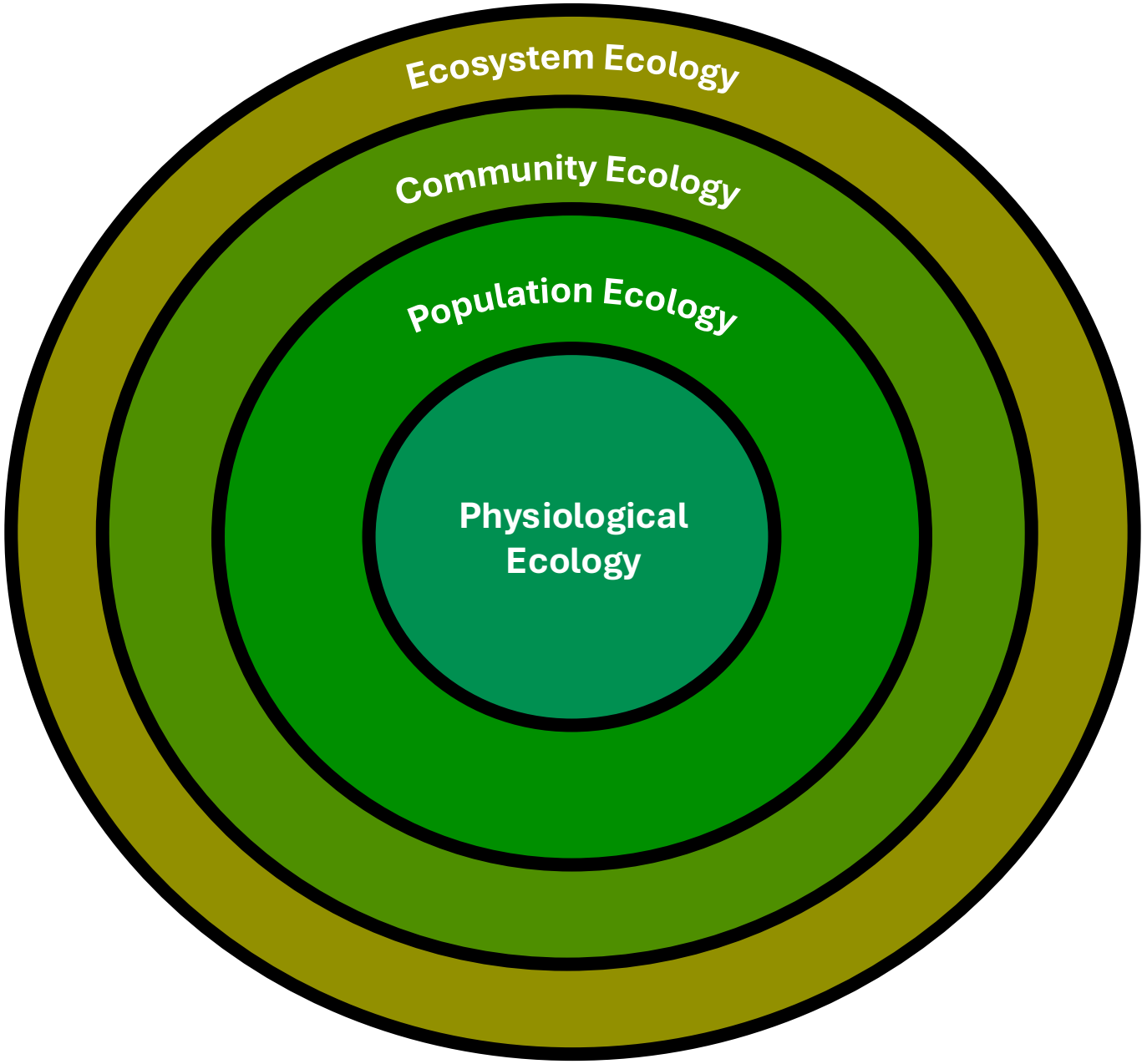


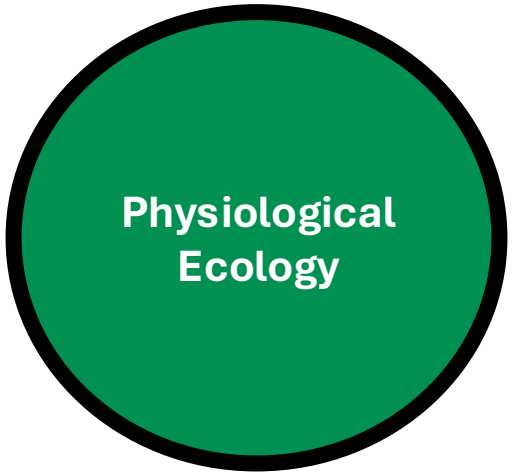
Physiology for Understanding and Knowledge in EcologY



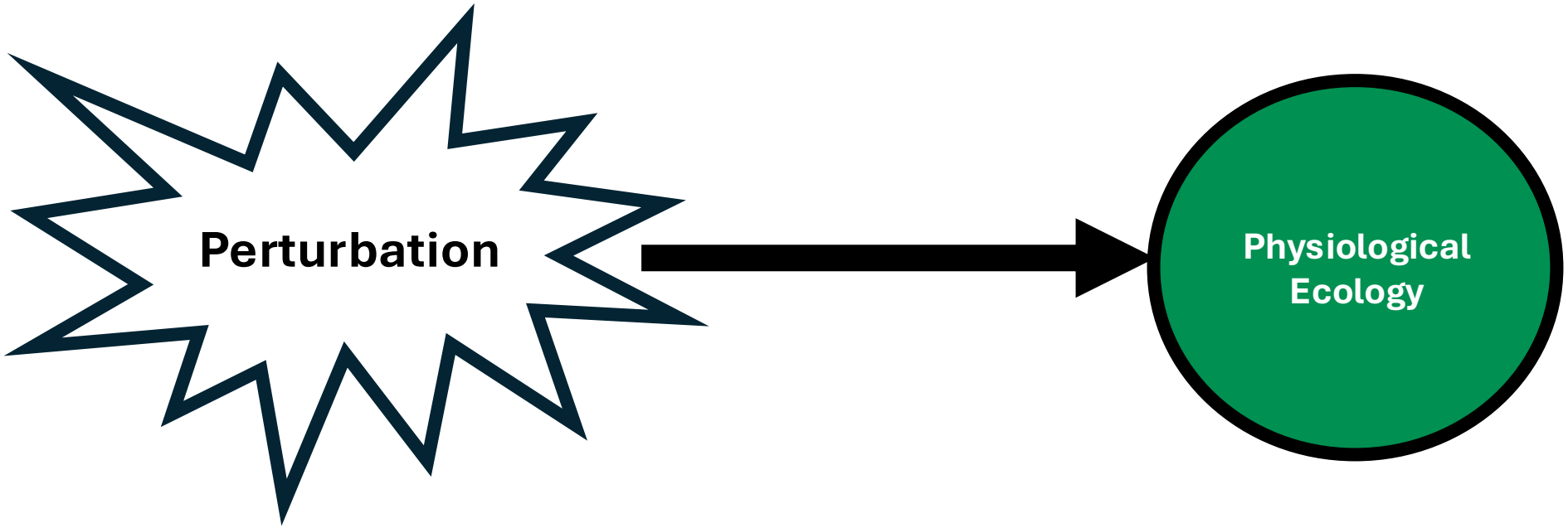
smithecophyslab.com

A primary theme of the lab is to use plant ecophysiology to better understand larger scale ecological processes

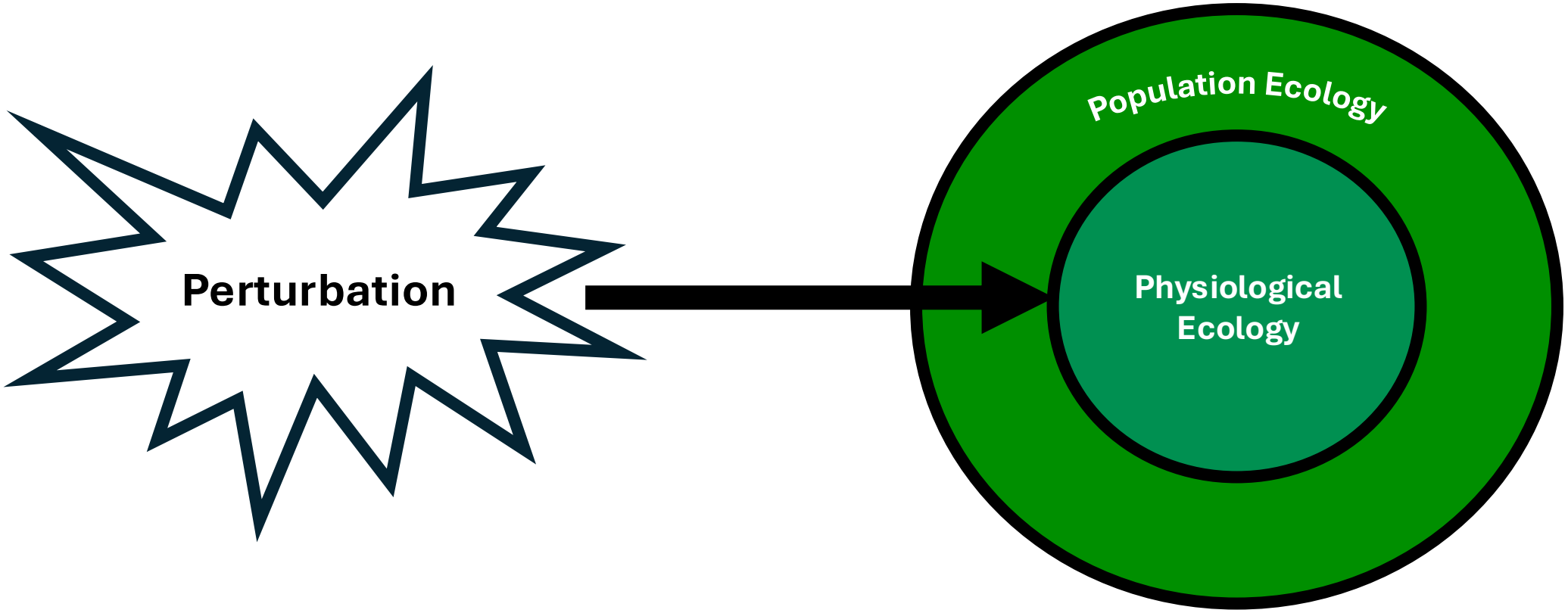




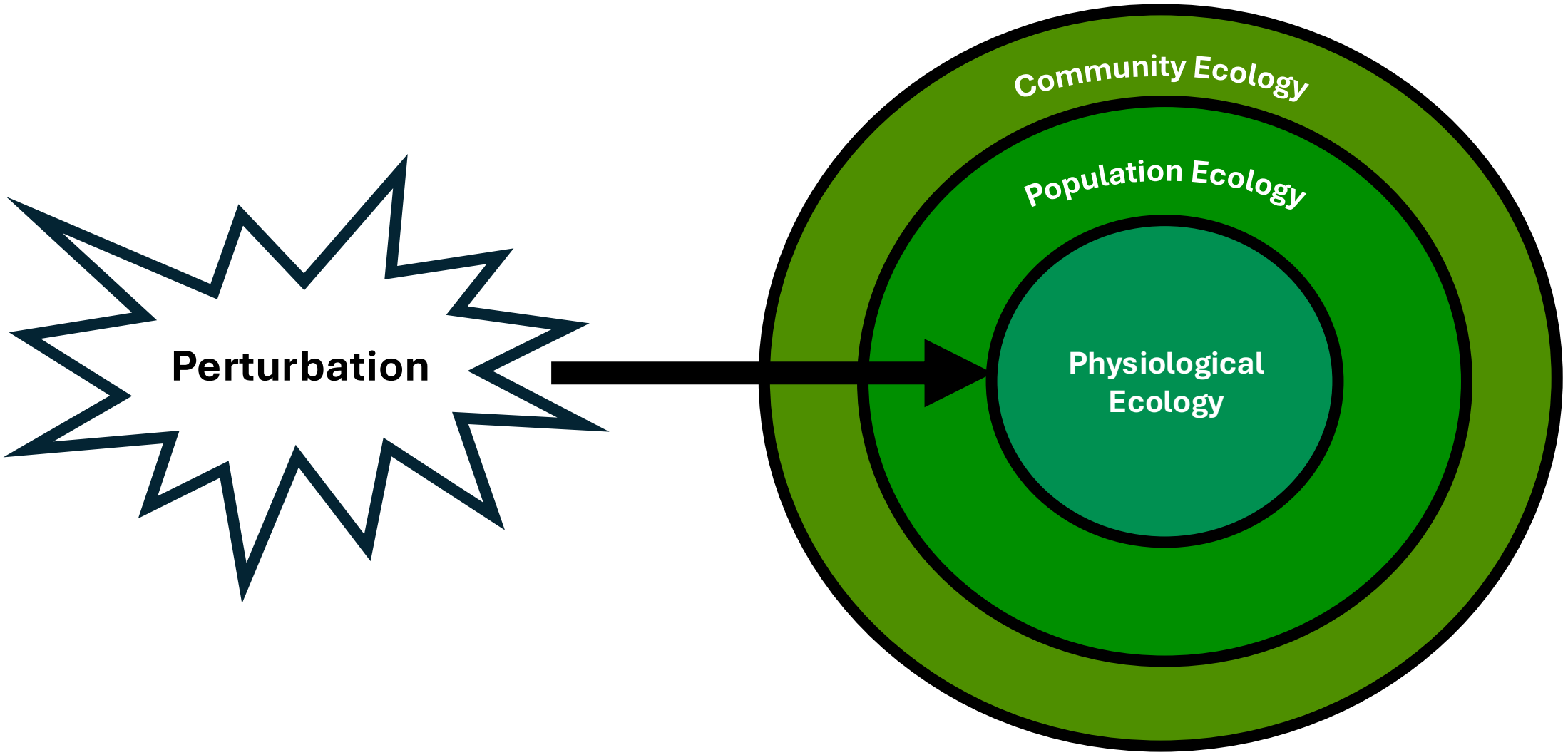
Smith (2024) *Am J Botany*



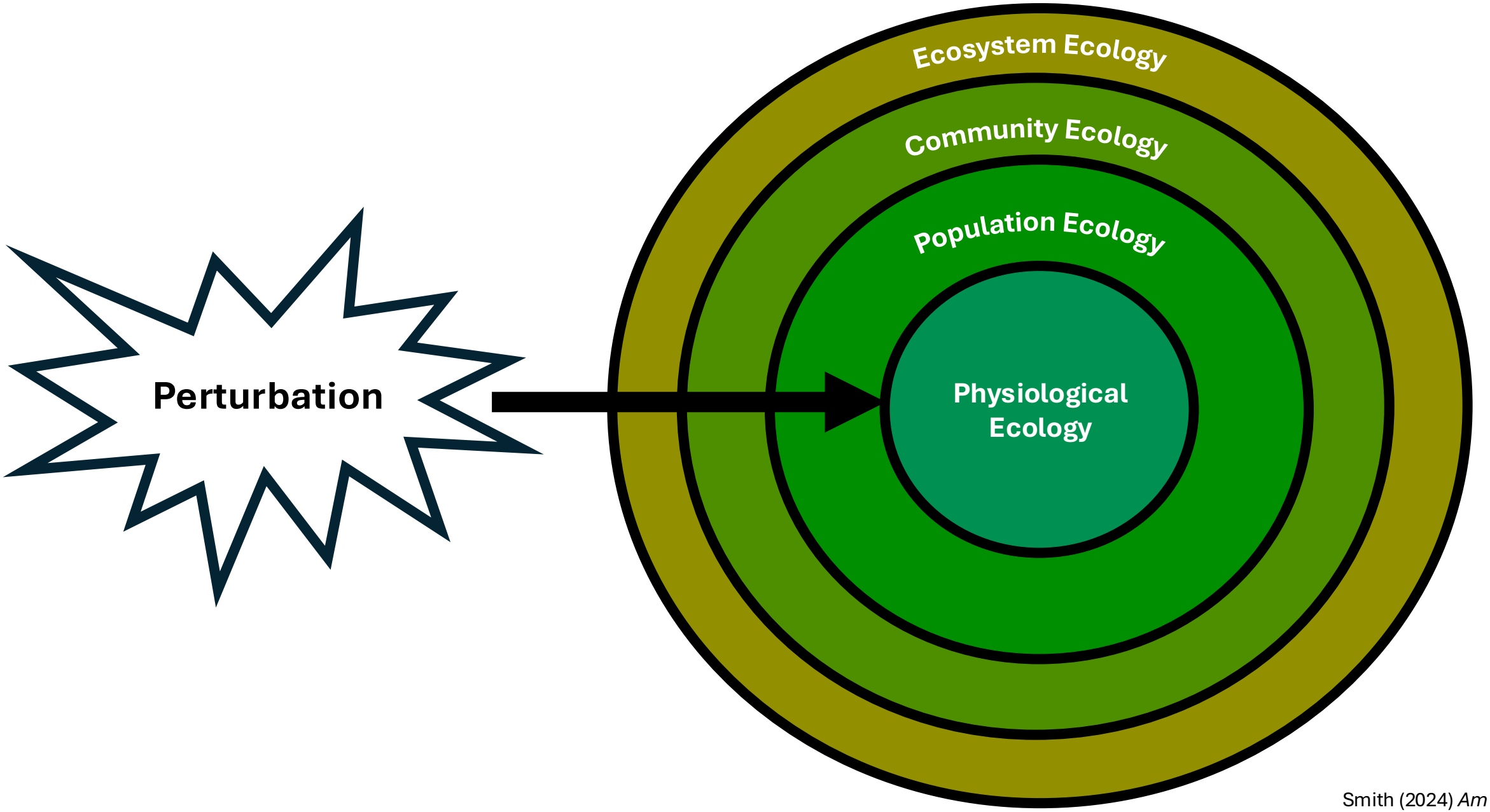
Smith (2024) *Am J Botany*



Smith (2024) *Am J Botany*



Smith (2024) *Am J Botany*



Smith (2024) *Am J Botany*

Structure of the talk

Structure of the talk

- **Short vignettes** illustrating connections between physiological and larger scale ecological processes under different **perturbations**

Structure of the talk

- **Short vignettes** illustrating connections between physiological and larger scale ecological processes under different **perturbations**
- Goal is to **highlight different questions and approaches**

Structure of the talk

- **Short vignettes** illustrating connections between physiological and larger scale ecological processes under different **perturbations**
- Goal is to **highlight different questions and approaches**
- Would be happy to discuss any in more detail after the talk

Approach icons

Manipulation experiments





Field observations

Theory development





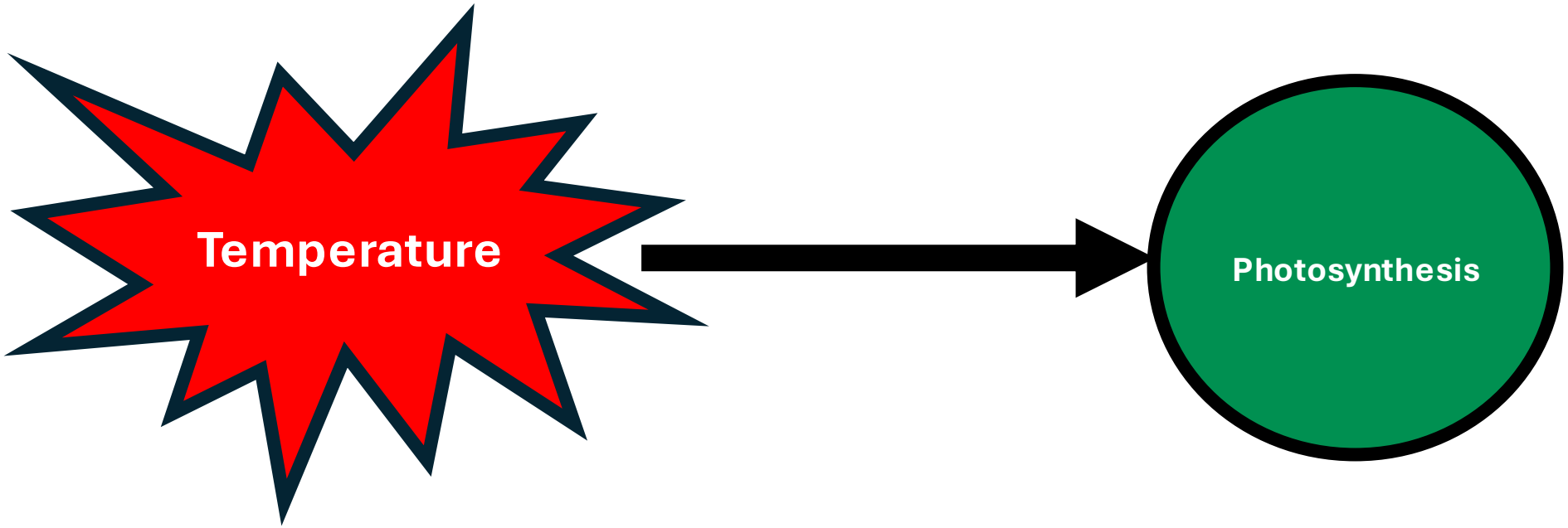
Process-based modeling

Perturbation vignettes

- **Temperature → global carbon and energy fluxes**
- **CO_2 → plant-nutrient dynamics**
- **Soil fertility → population dynamics**
- **Environmental heterogeneity → community dynamics**
- **Invasion → mycorrhizal symbioses**

Perturbation vignettes

- **Temperature** → global carbon and energy fluxes
- **CO_2** → soil nutrient dynamics
- **Soil fertility** → **population dynamics**
- **Environmental heterogeneity** → **community dynamics**
- **Invasion** → mycorrhizal symbioses


Temperature
Photosynthesis

Developed theory on temperature acclimation of photosynthesis and tested it with data

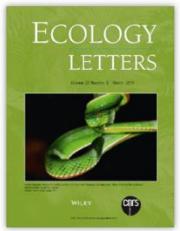
ECOLOGY LETTERS

Letter | Open Access

Global photosynthetic capacity is optimized to the environment

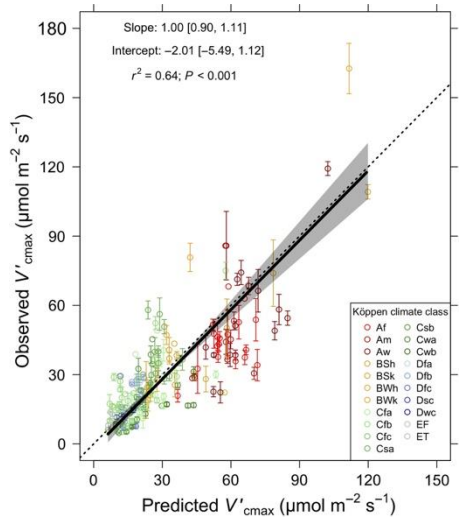
Ecology Letters homepage

Nicholas G. Smith, Trevor F. Keenan, I. Colin Prentice, Han Wang, Ian J. Wright, Ülo Niinemets, Kristine Y. Crous, Tomas F. Domingues, Rossella Guerrieri, F. Yoko Ishida, Jens Kattge ... See all authors



Volume 22, Issue 3
March 2019
Pages 506-517

Figures | References | Related | Information



Global Change Biology

Global Change Biology homepage

PRIMARY RESEARCH ARTICLE | Full Access

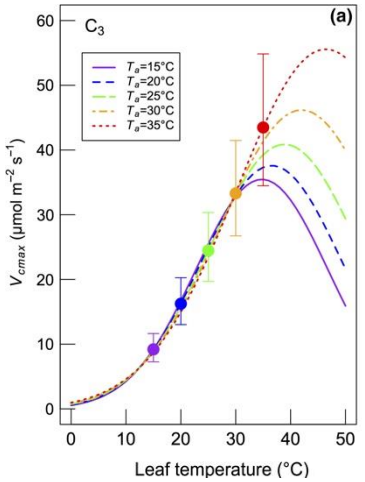
Short-term acclimation to warmer temperatures accelerates leaf carbon exchange processes across plant types

Nicholas G. Smith, Jeffrey S. Dukes

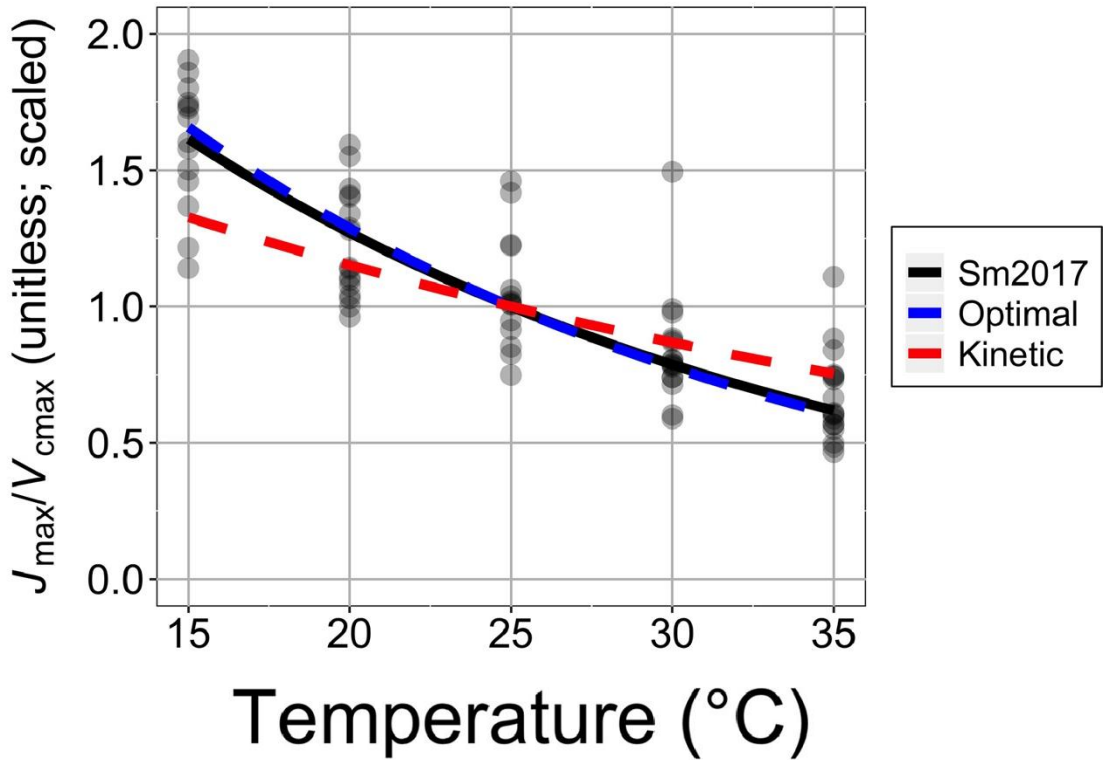


Volume 23, Issue 11
November 2017
Pages 4840-4853

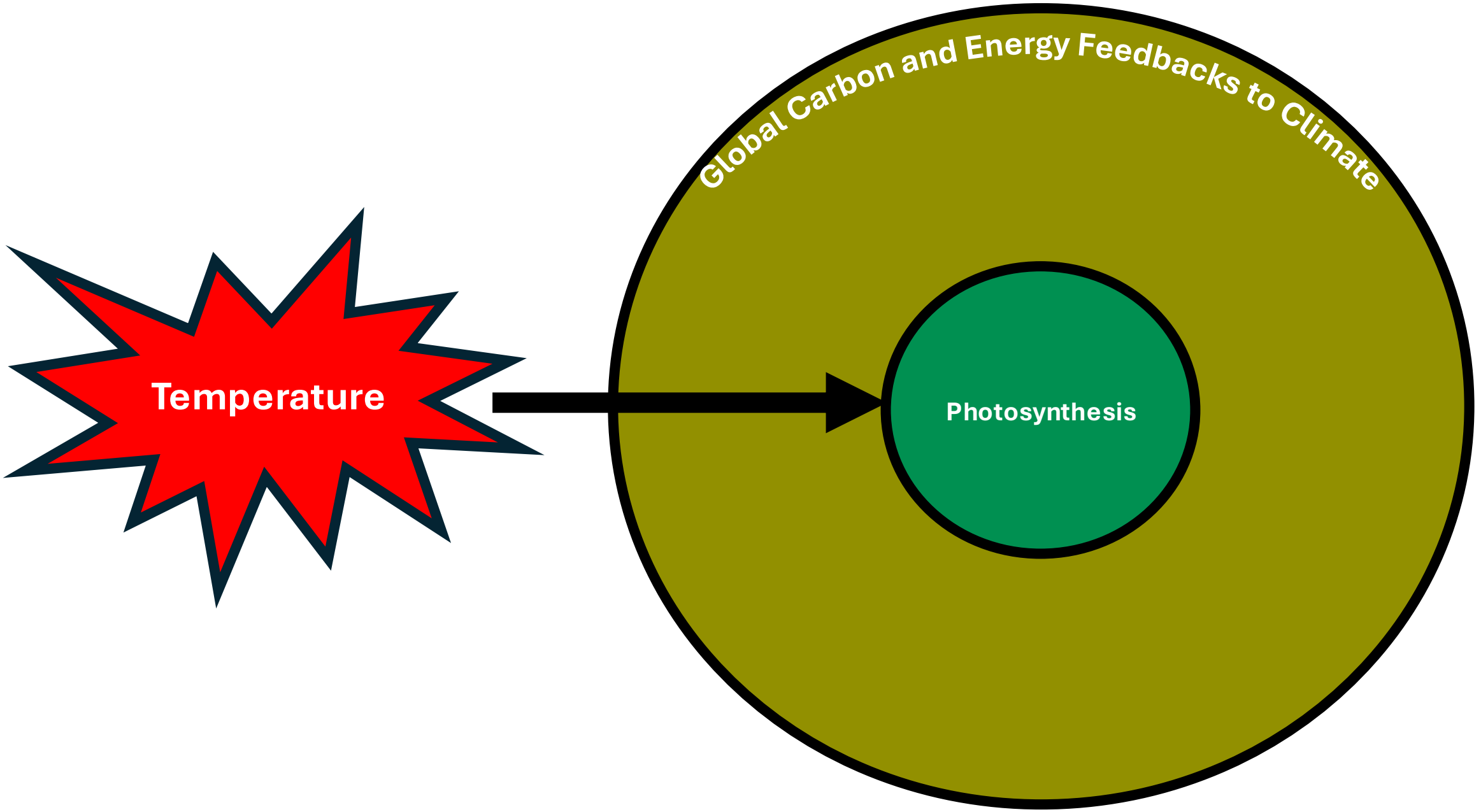
Figures | References | Related | Information



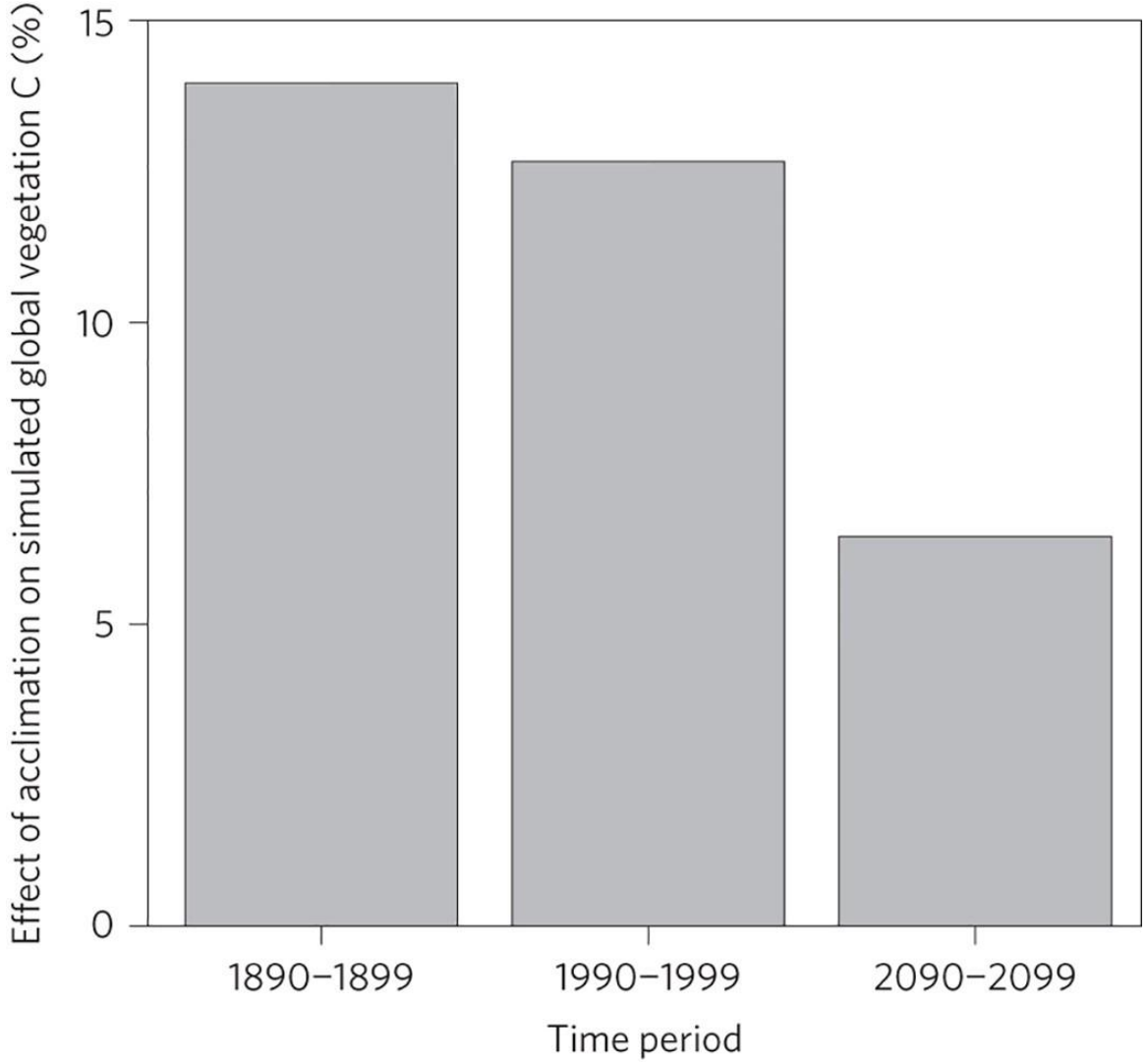
Photosynthesis shows strong and predictable acclimation responses to temperature



Smith and Keenan (2020) *Glob Change Bio*

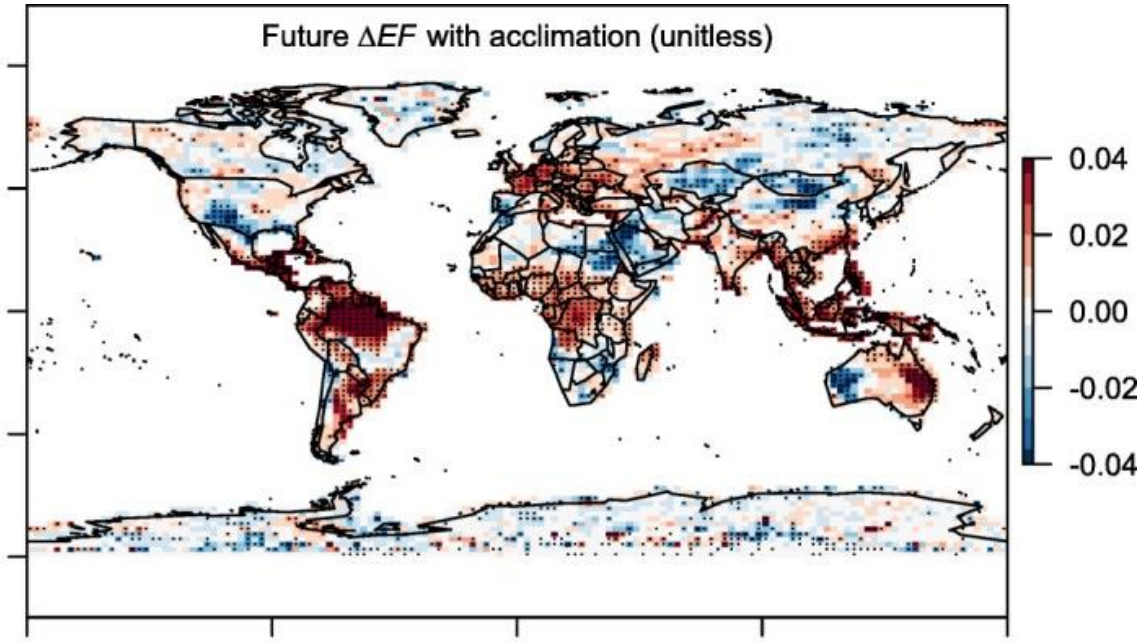

Global Carbon and Energy Feedbacks to Climate
Temperature
Photosynthesis

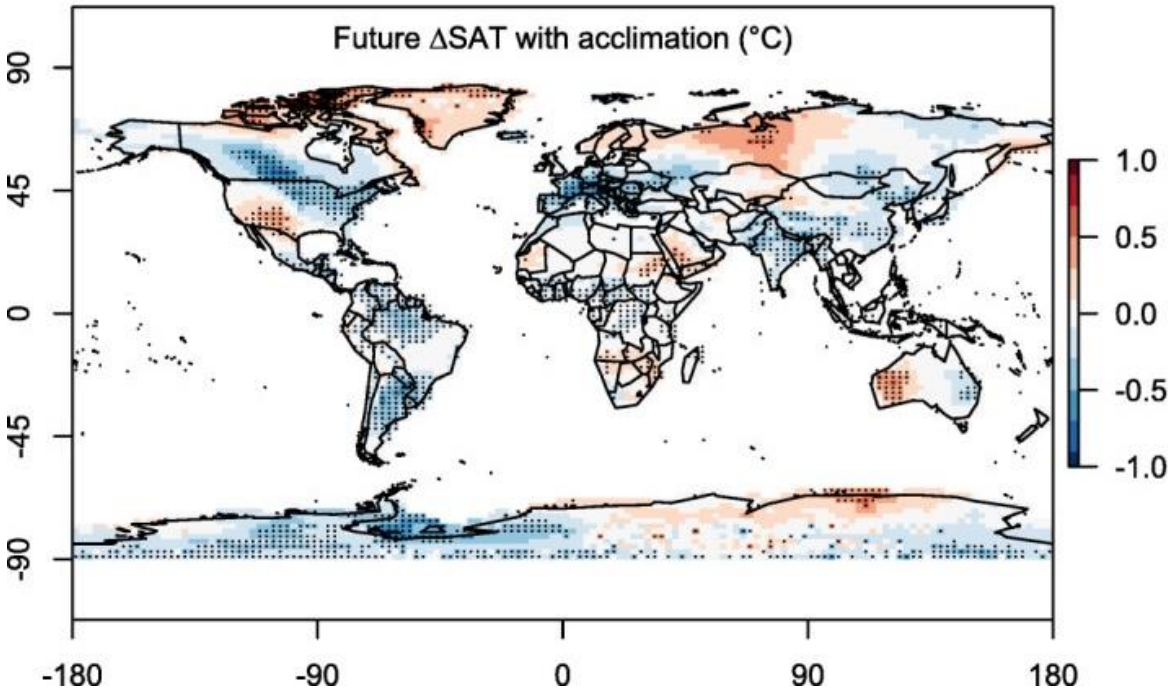
Photosynthetic temperature acclimation increases simulated global vegetation carbon



Smith et al. (2016) *Nat Clim Change*

Photosynthetic temperature acclimation changes transpiration and air temperature

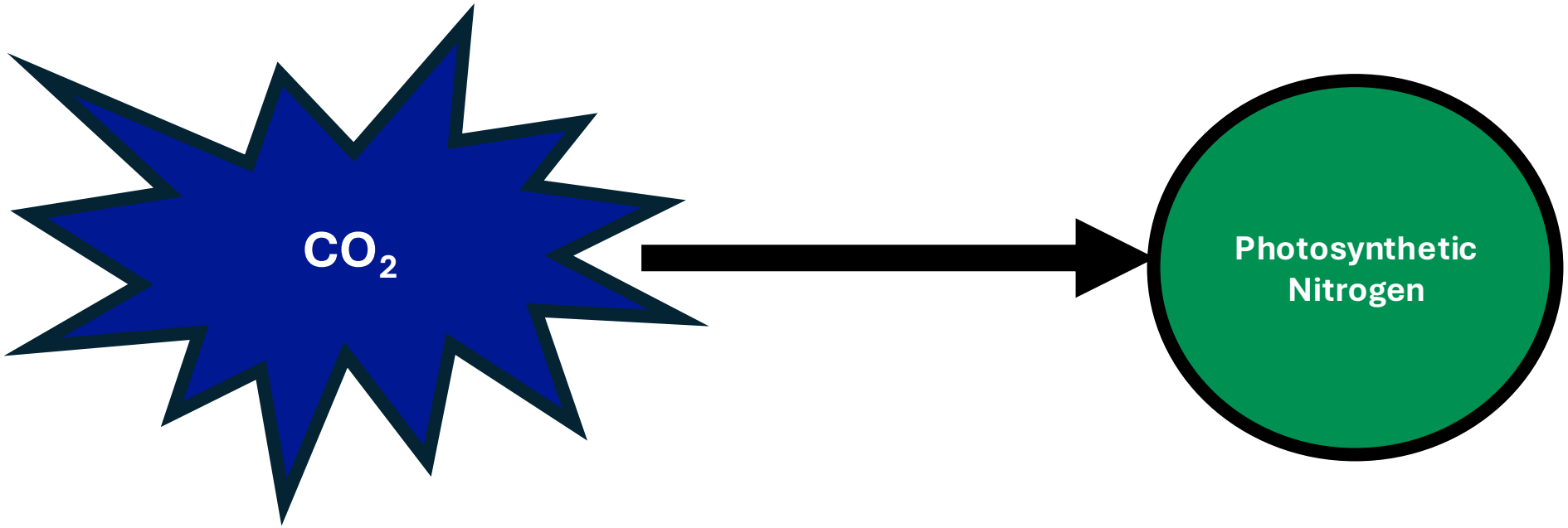




Smith et al. (2017) *JAMES*

Perturbation vignettes

- Temperature → global carbon and energy fluxes
- **CO_2 → plant-nutrient dynamics**
- Soil fertility → population dynamics
- Environmental heterogeneity → community dynamics
- Invasion → mycorrhizal symbioses


CO_2
Photosynthetic
Nitrogen

Theory suggests that downregulation of photosynthetic N is an acclimation, rather than nutrient limitation response

Global Change Biology

PRIMARY RESEARCH ARTICLE | Full Access

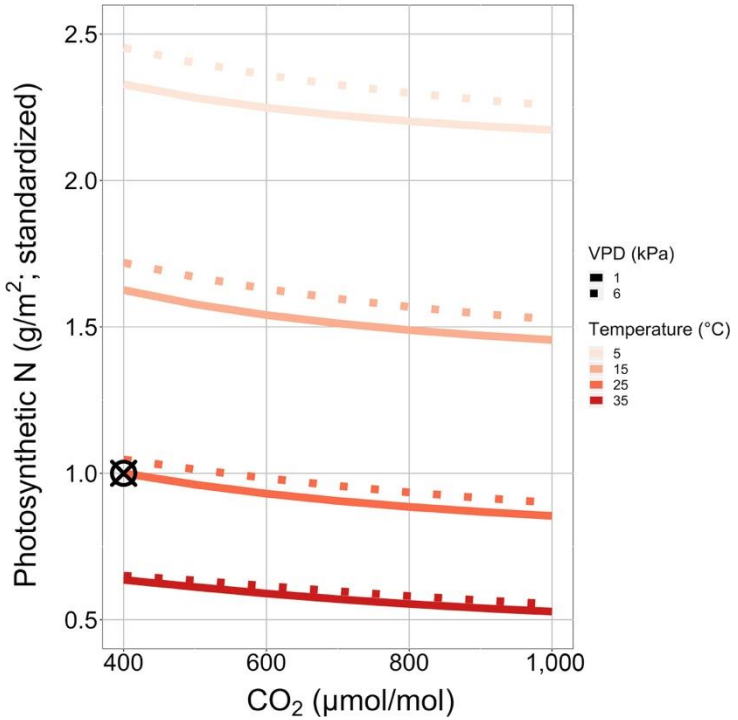
Mechanisms underlying leaf photosynthetic acclimation to warming and elevated CO_2 as inferred from least-cost optimality theory

Nicholas G. Smith, Trevor F. Keenan



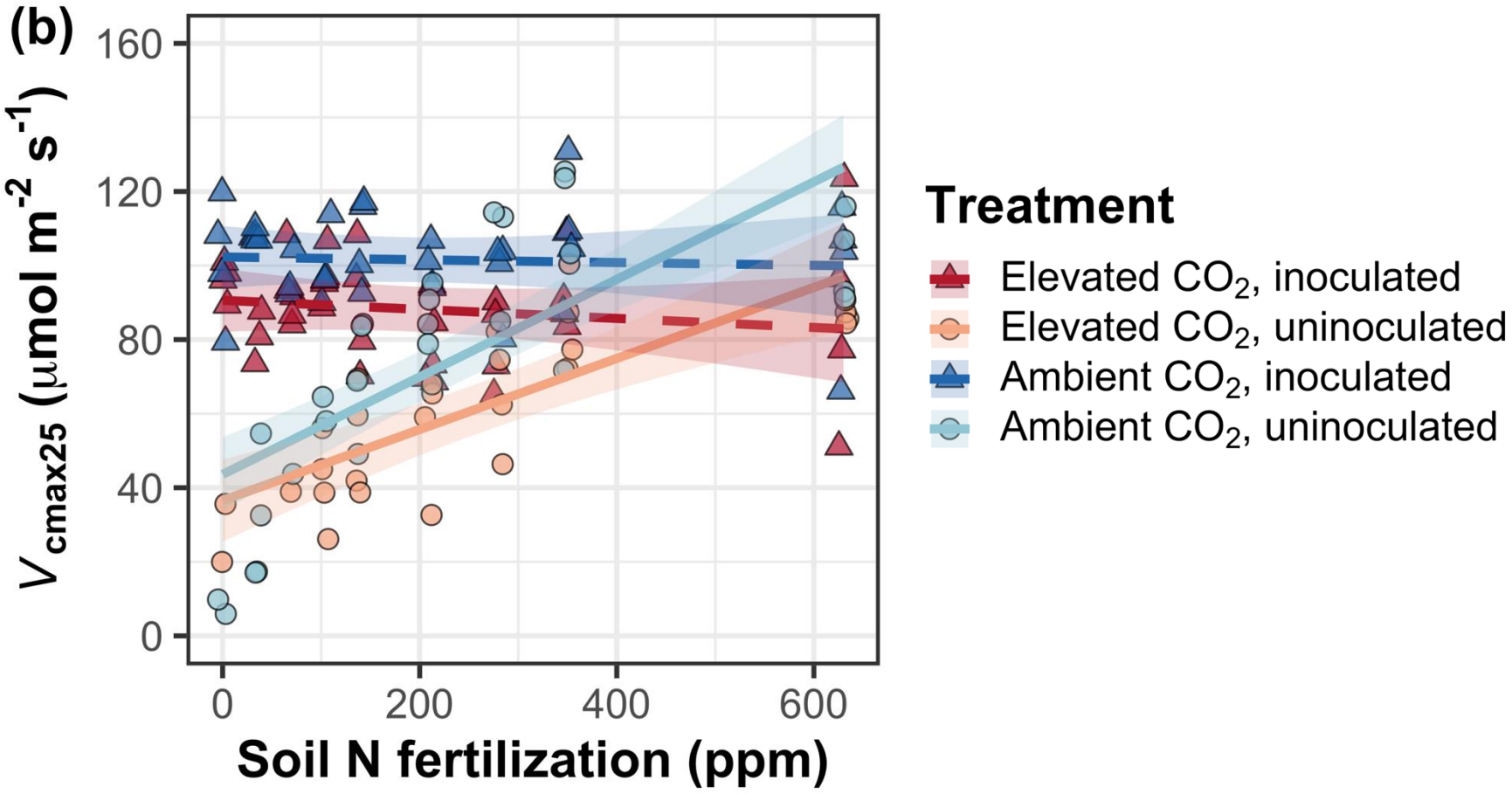
Volume 26, Issue 9
September 2020
Pages 5202-5216

Figures | References | Related | Information

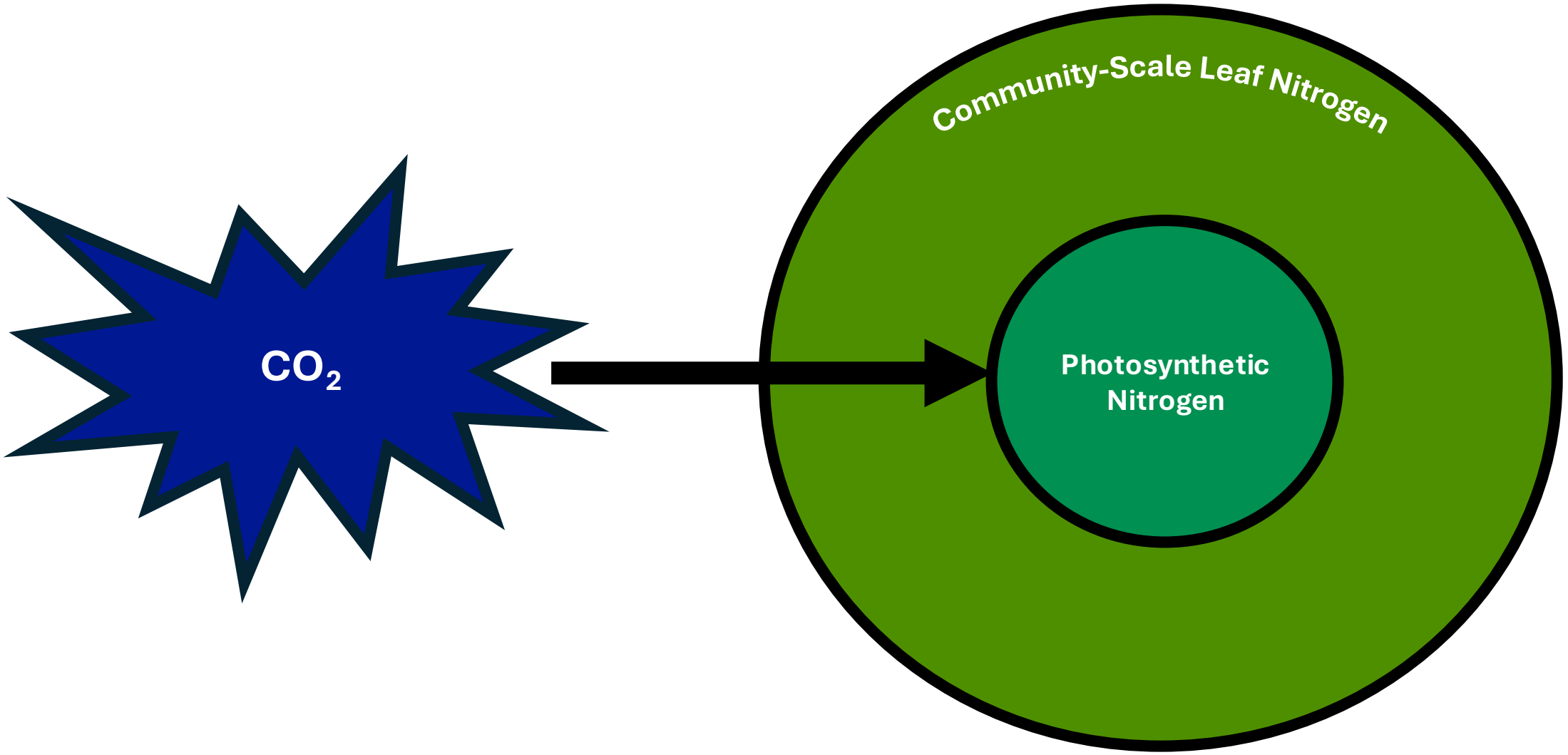




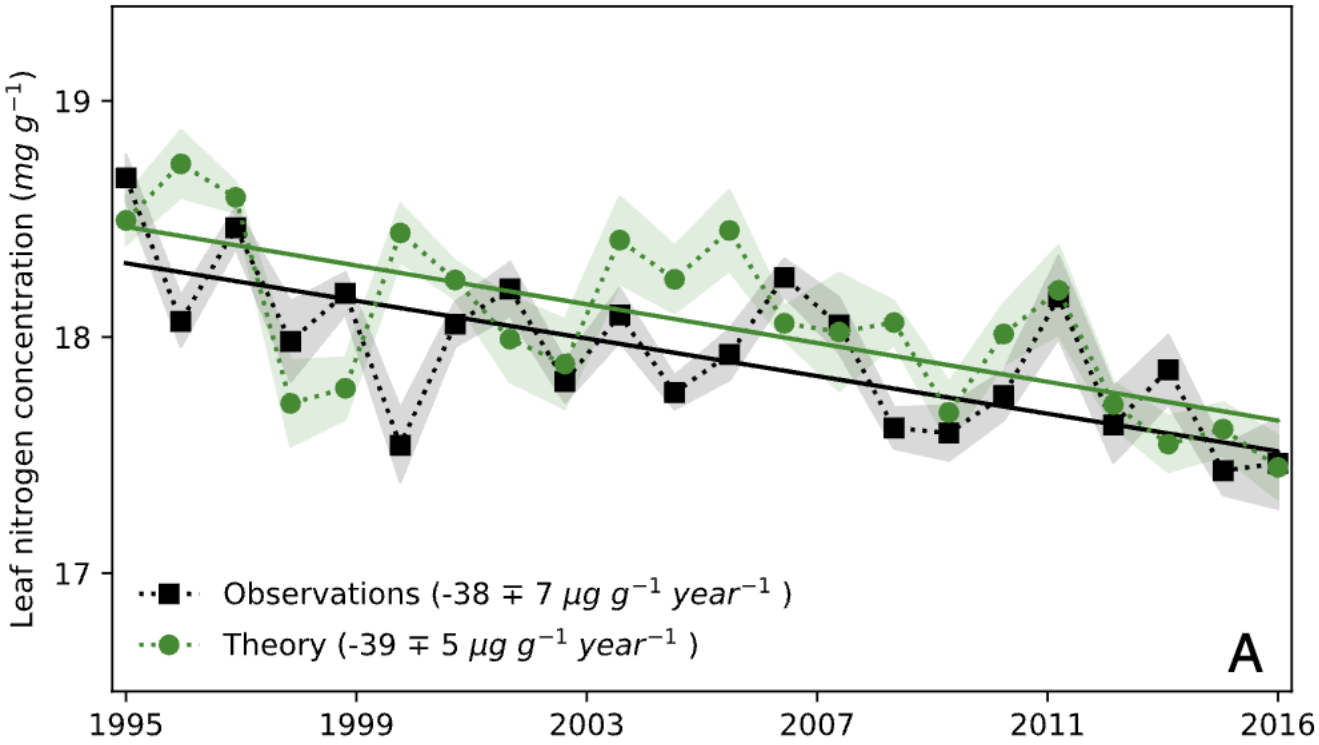
Elevated CO_2 reduces leaf nitrogen demand regardless of nitrogen availability



Perkowski, Ezekannagha, Smith (2025) *J Exp Bot*


CO_2
Community-Scale Leaf Nitrogen
Photosynthetic Nitrogen

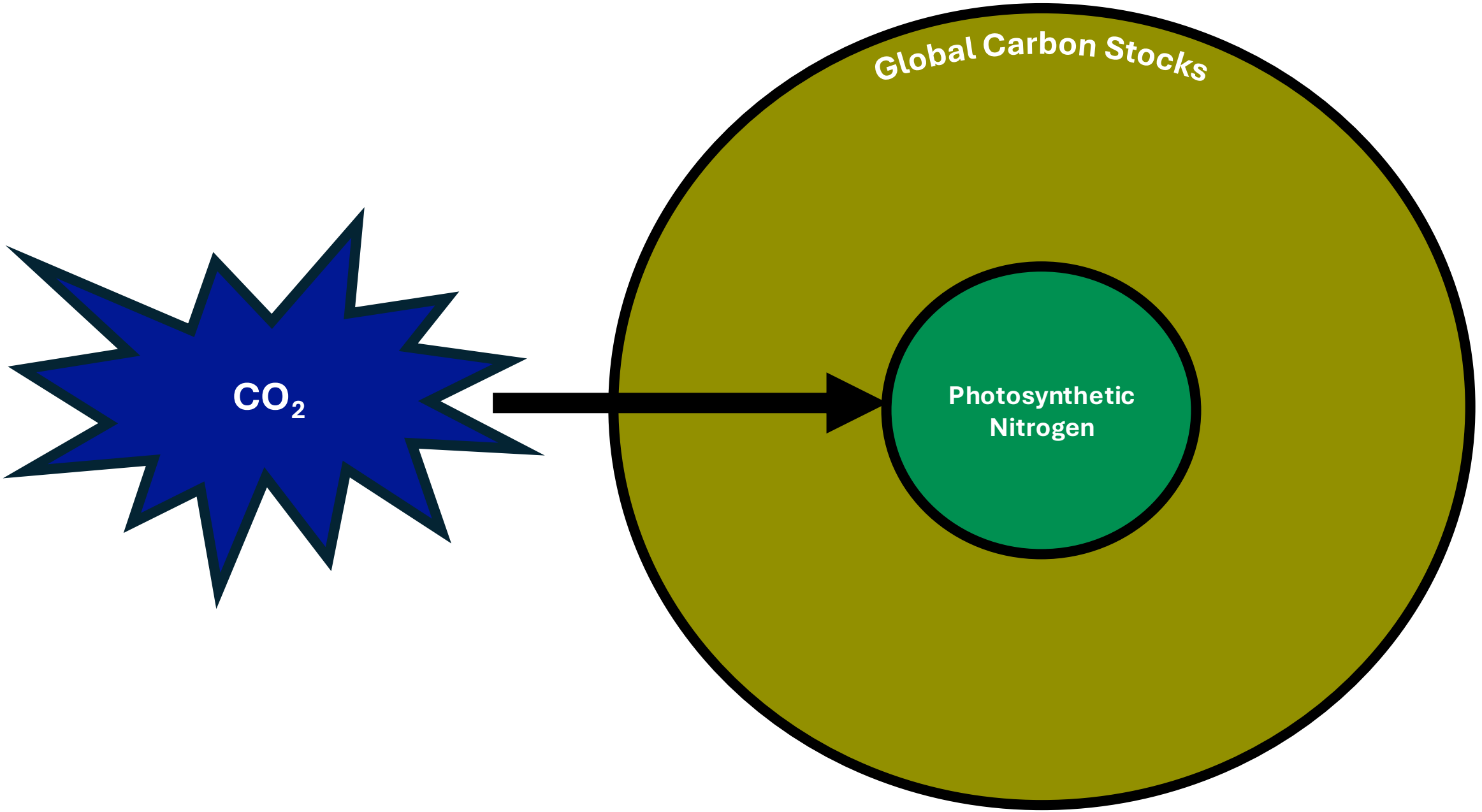
This is predictable over continental gradients



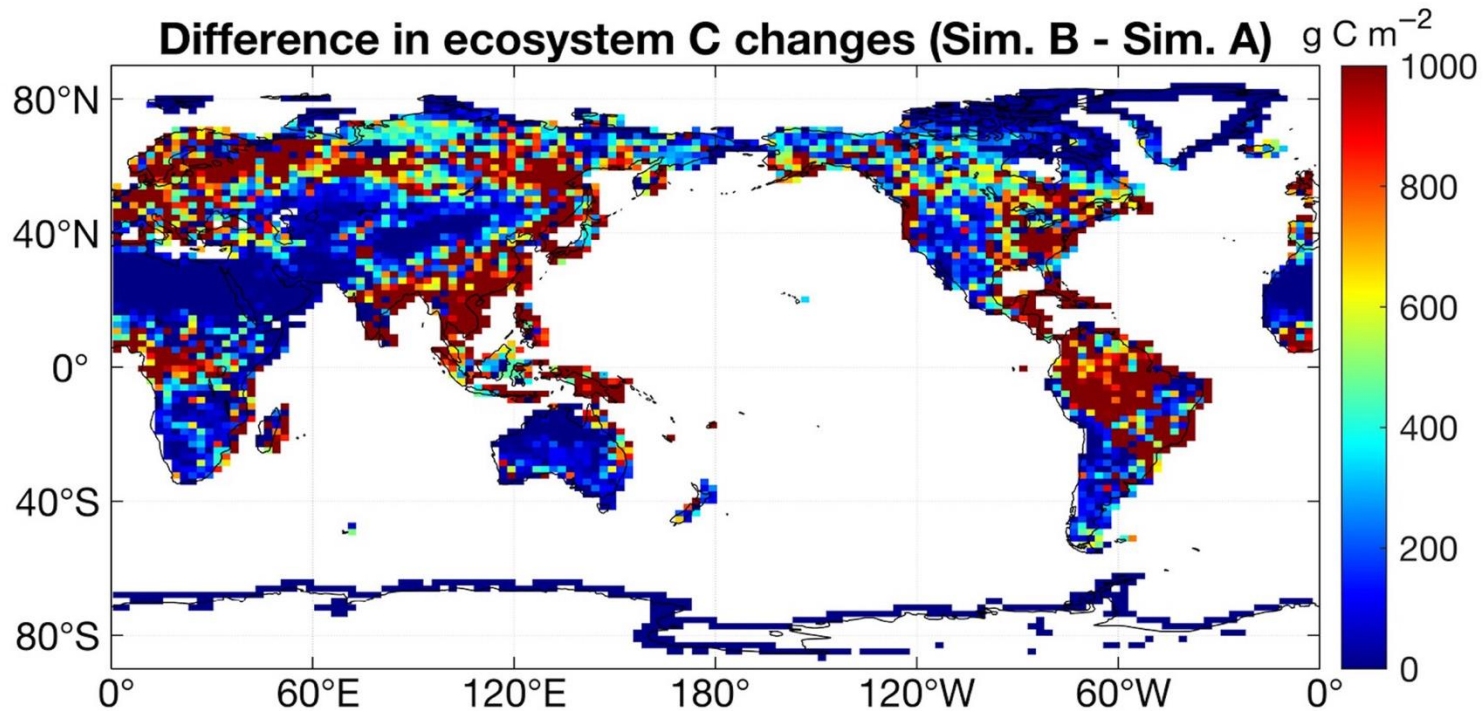




Bassiouni, Smith, et al. (2025) *PNAS*

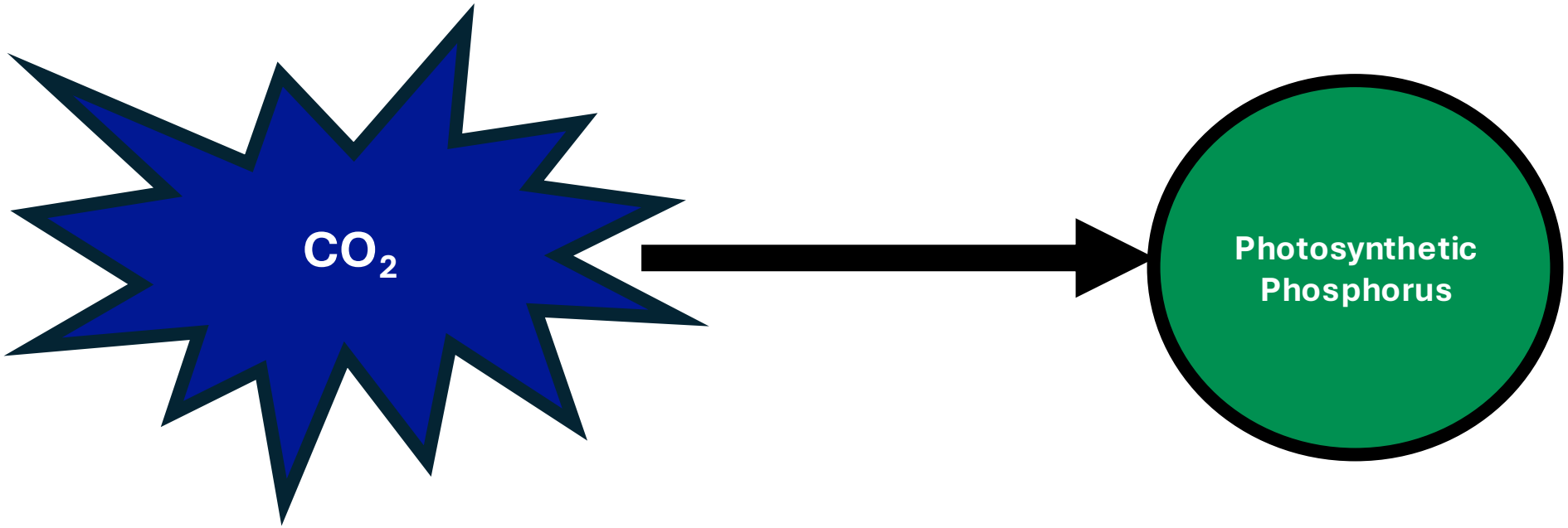

Global Carbon Stocks
CO_2
Photosynthetic Nitrogen

Accounting for this effect in models increases simulated future global carbon stocks

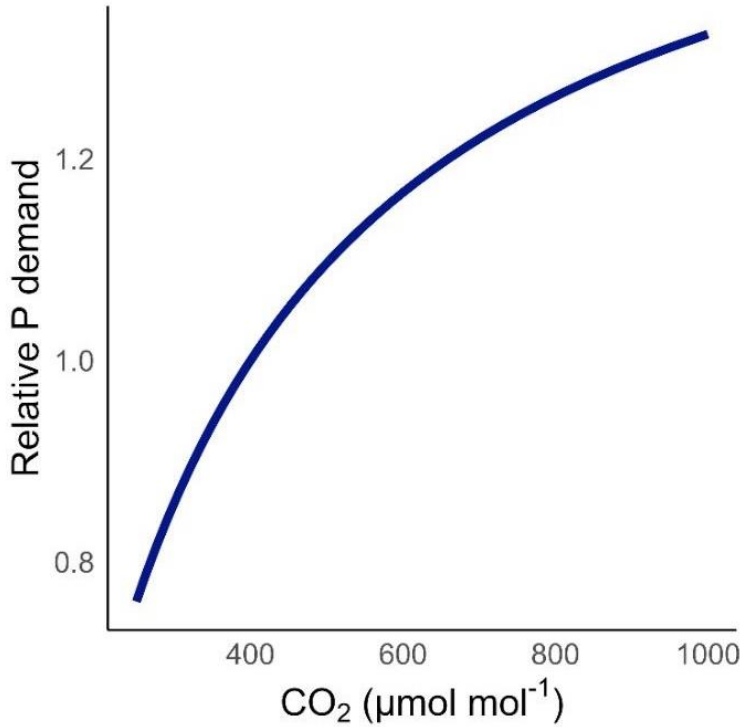


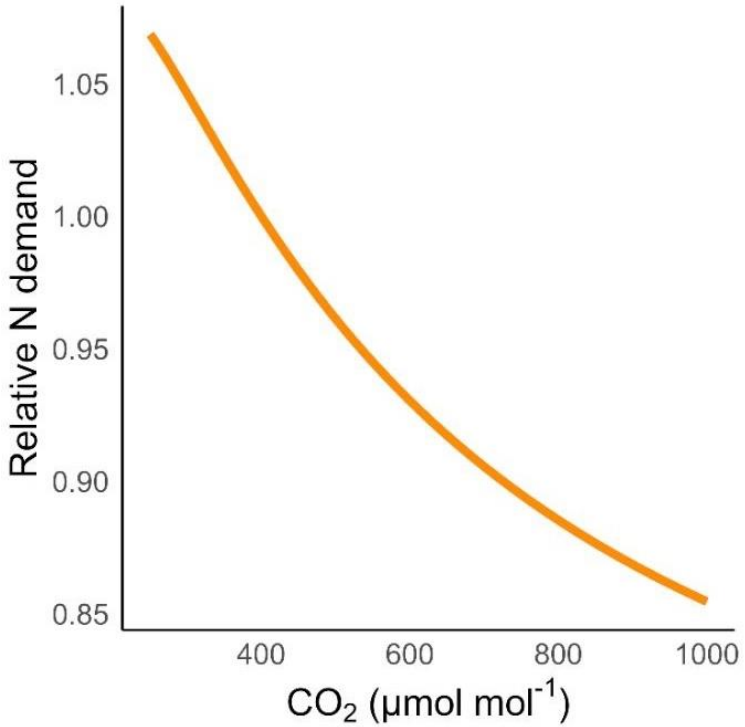


Smith et al. (2024) *Glob Change Biol*


CO_2
Photosynthetic
Phosphorus

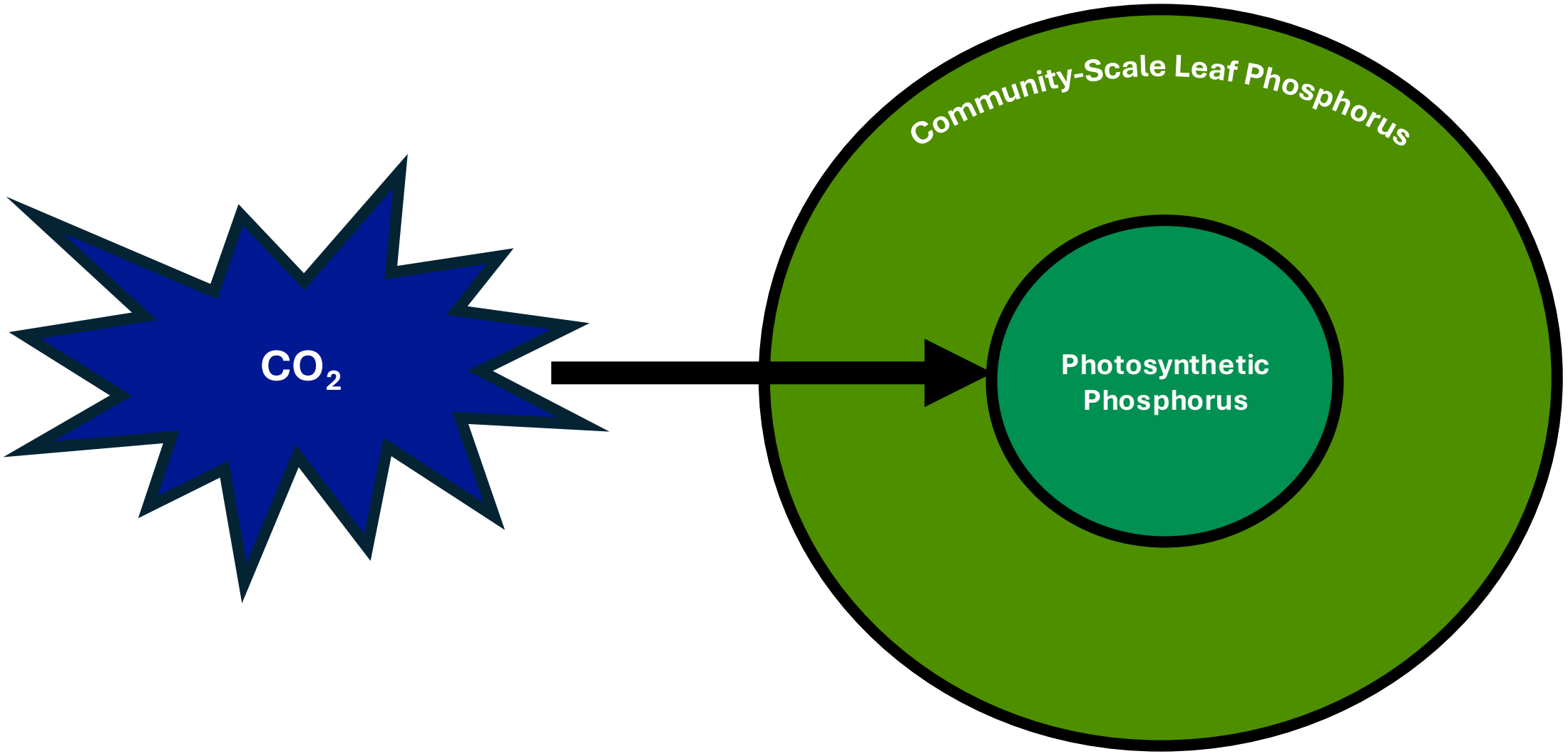
Phosphorus demand response differs from nitrogen demand response to CO_2



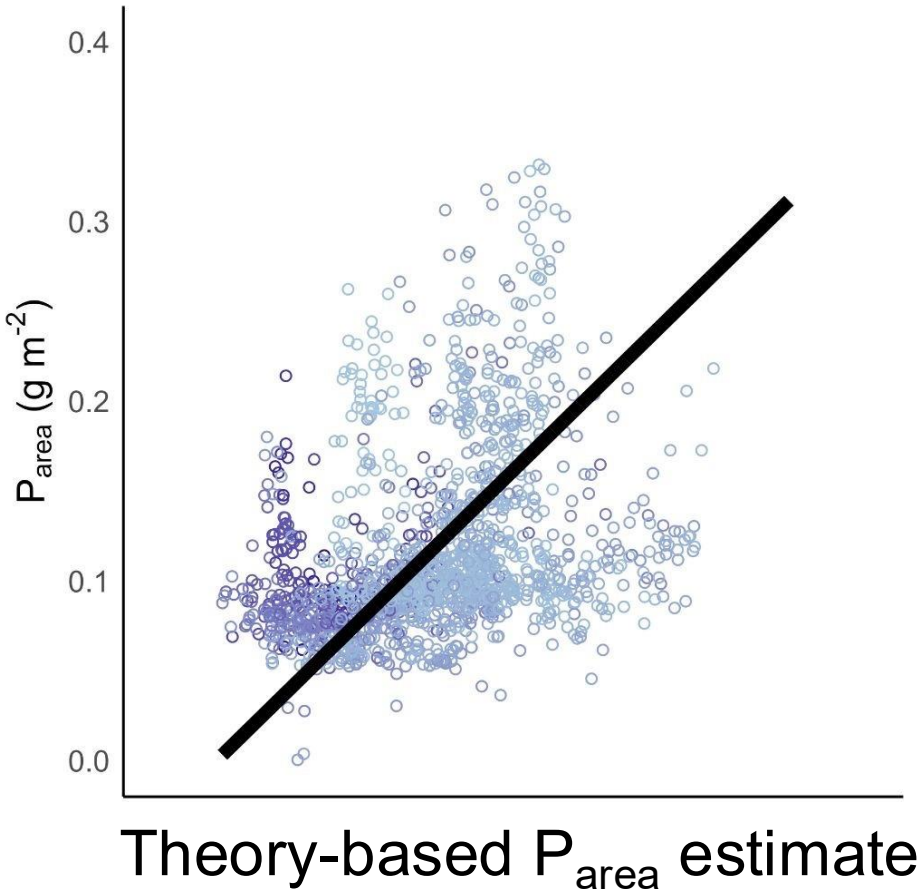




Smith et al. (In Prep)


CO_2
Community-Scale Leaf Phosphorus
Photosynthetic Phosphorus

Theory can predict community-scale leaf phosphorus



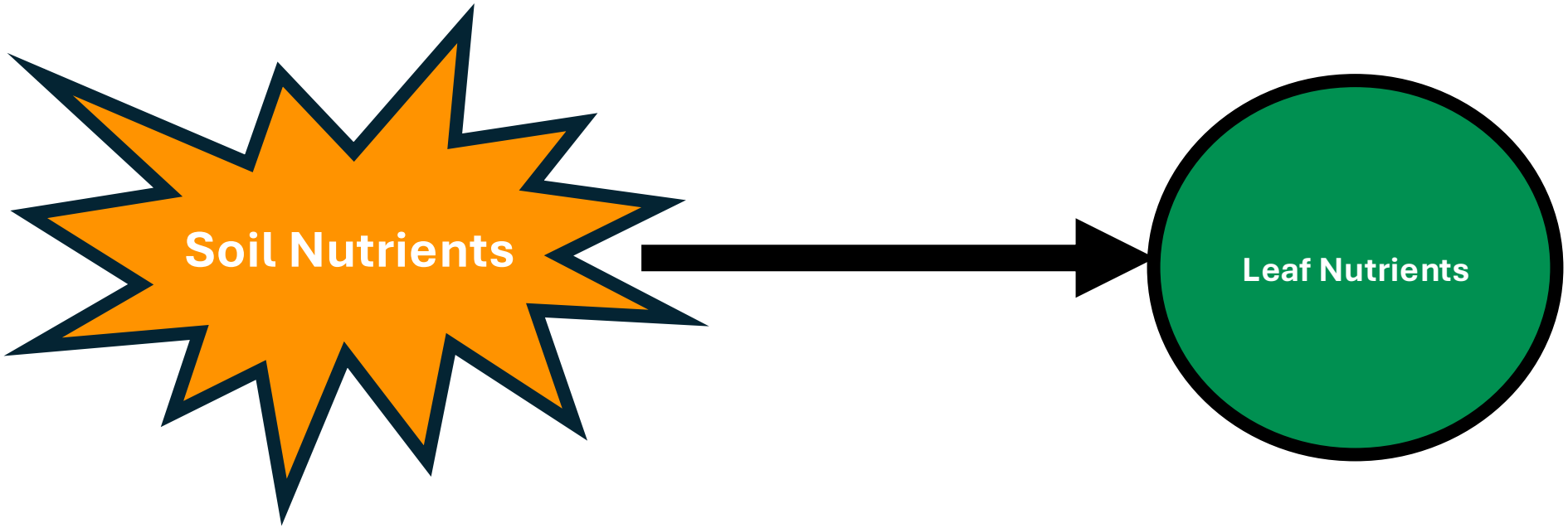




Smith et al. (In Prep)

Perturbation vignettes

- Temperature → global carbon and energy fluxes
- CO_2 → plant-nutrient dynamics
- **Soil fertility → population dynamics**
- Environmental heterogeneity → community dynamics
- Invasion → mycorrhizal symbioses


Soil Nutrients
Leaf Nutrients

Climate can change theoretical photosynthetic demand for nutrients

Global Change Biology

PRIMARY RESEARCH ARTICLE | Full Access

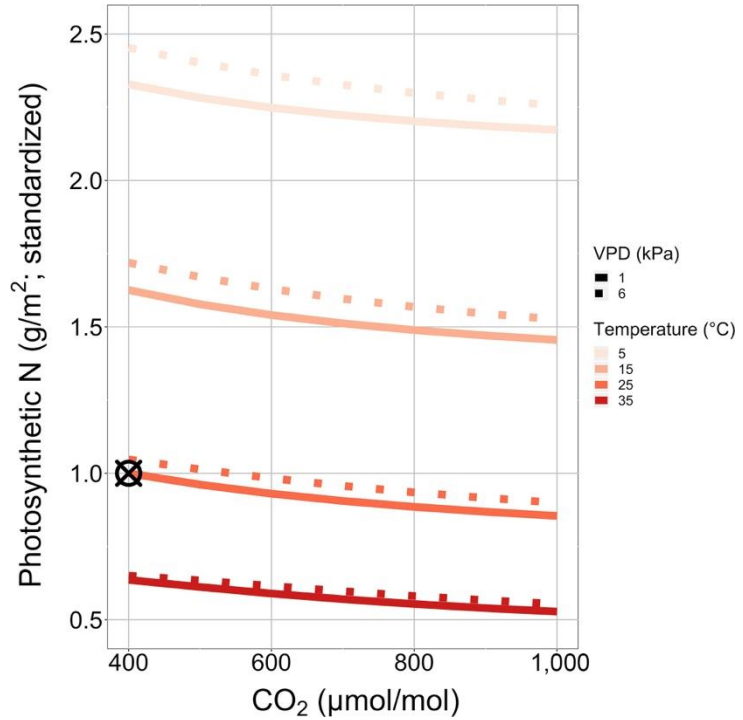
Mechanisms underlying leaf photosynthetic acclimation to warming and elevated CO_2 as inferred from least-cost optimality theory

Nicholas G. Smith, Trevor F. Keenan



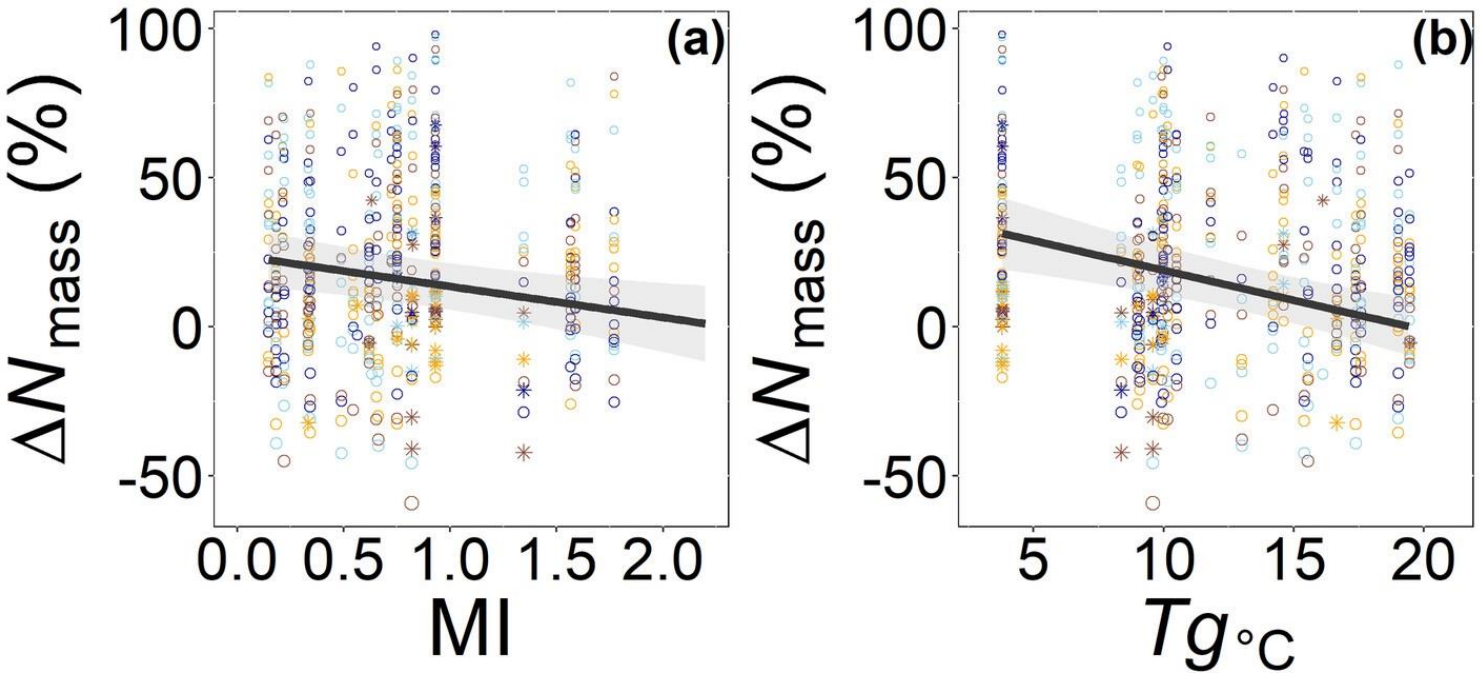
Volume 26, Issue 9
September 2020
Pages 5202-5216

Figures References Related Information

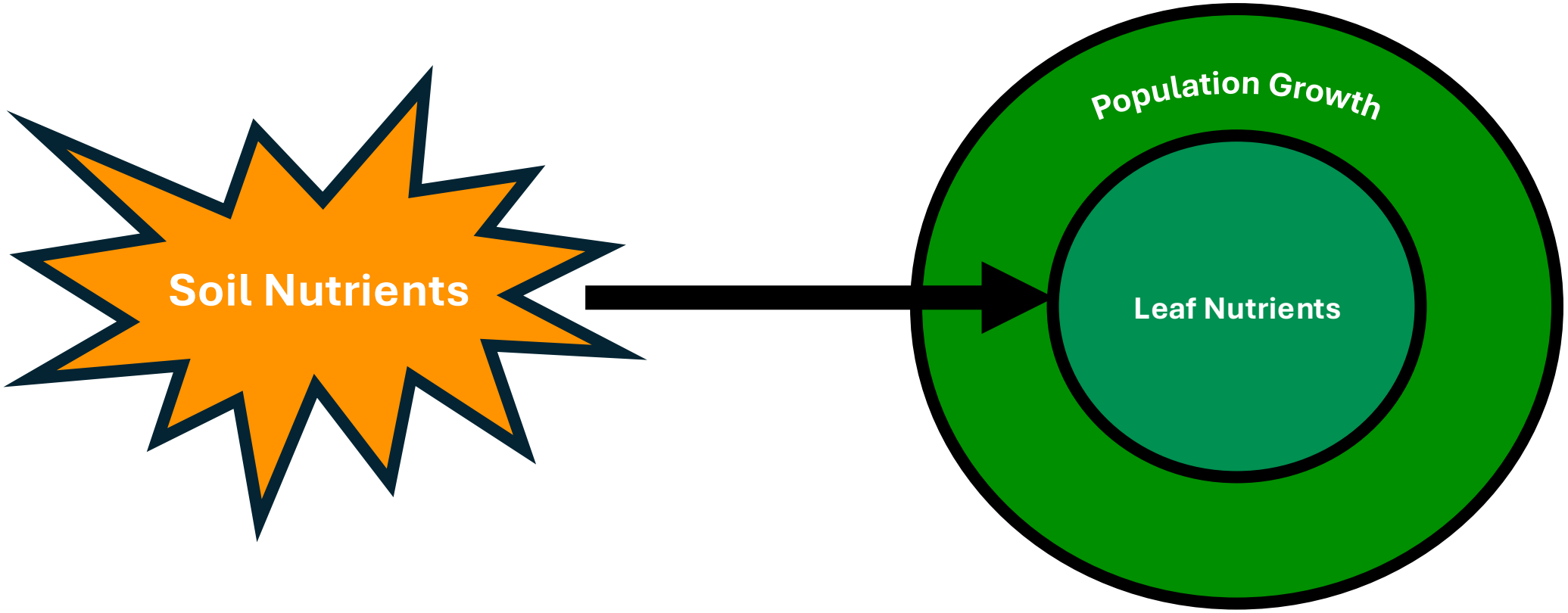




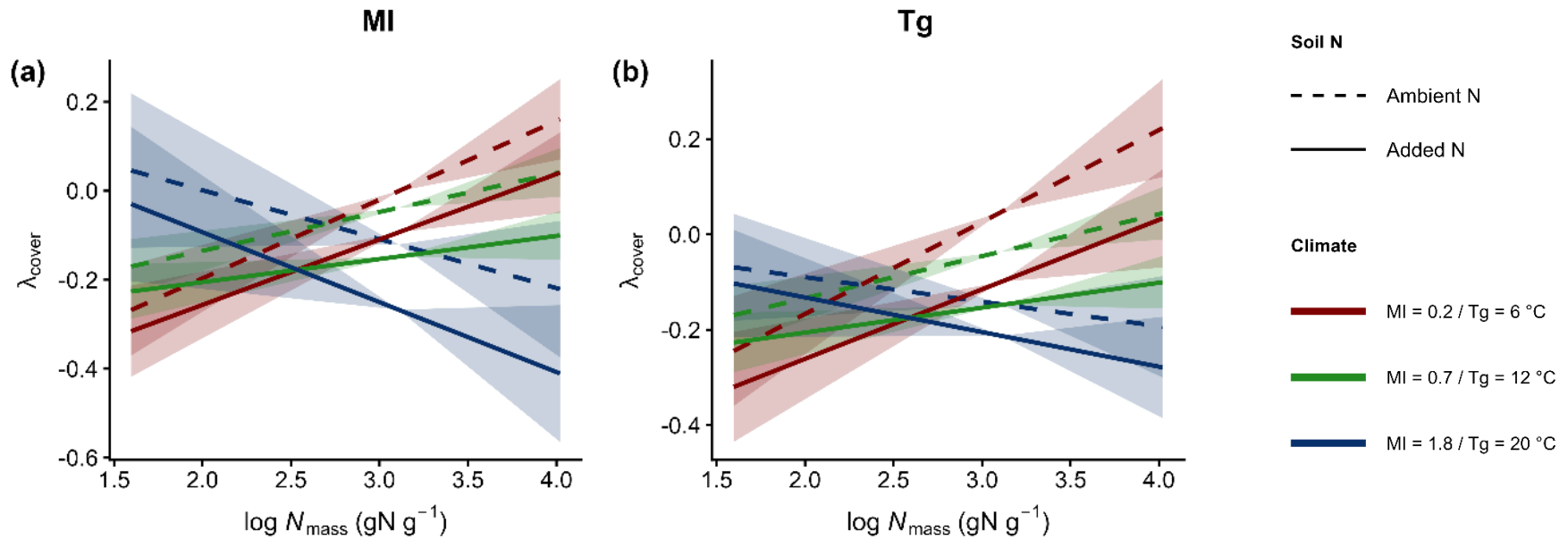
Soil nutrients enhance leaf nutrients where photosynthetic demand is greatest (dry/cold)



Cheaib…and Smith (2025) *Ecol Letters*


Soil Nutrients
Population Growth
Leaf Nutrients

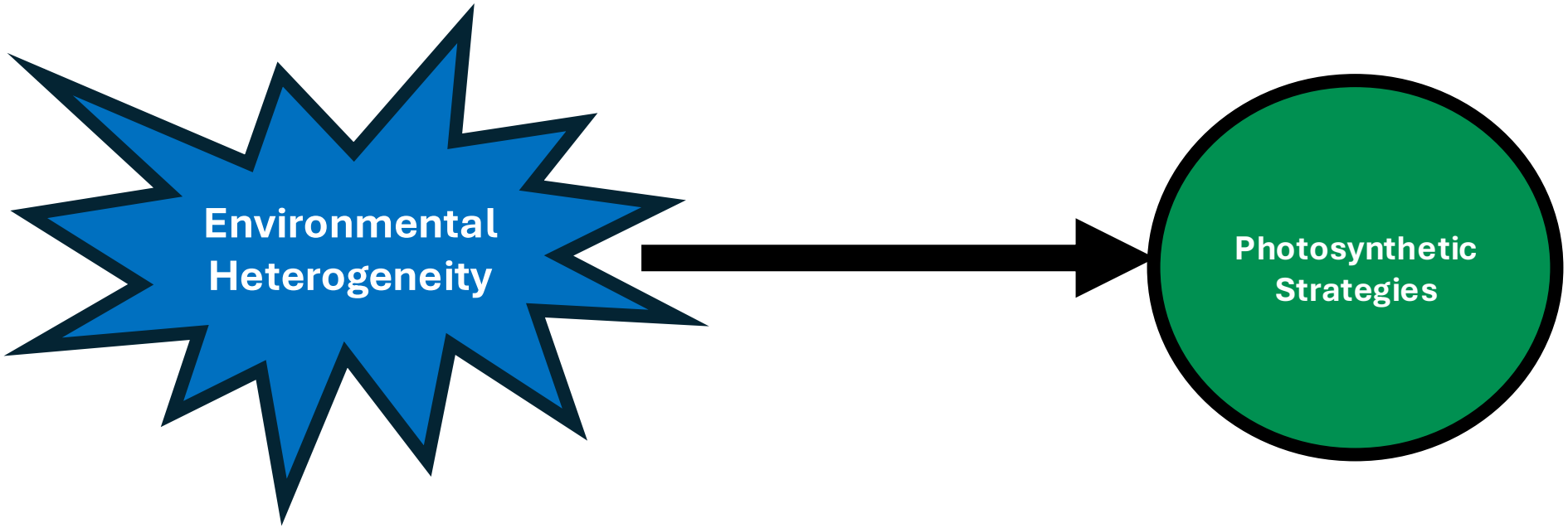
In cold/dry places, higher leaf nitrogen confers greater species population growth



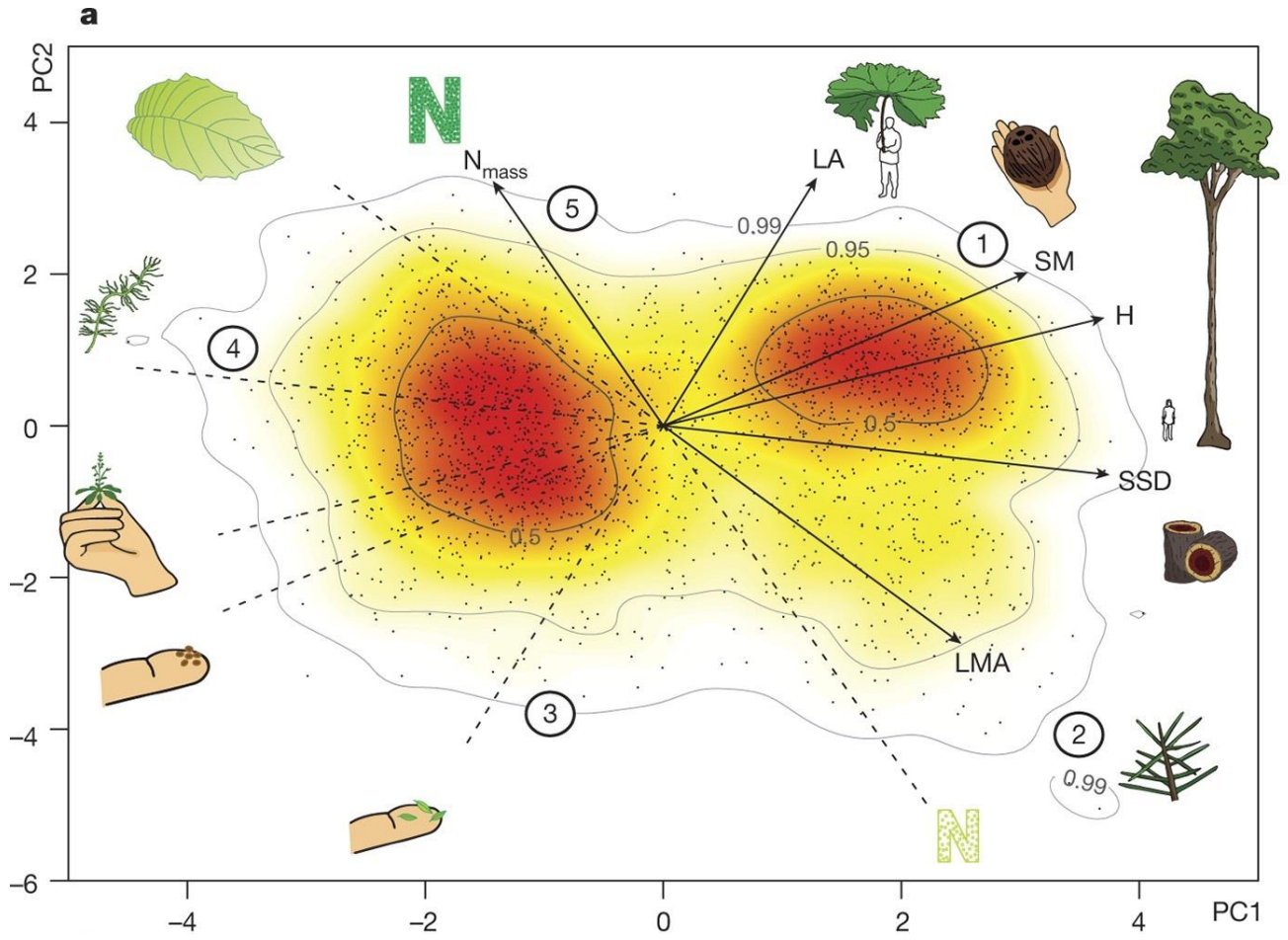
Cheaib…and Smith (in Prep)

Perturbation vignettes

- Temperature → global carbon and energy fluxes
- CO_2 → plant-nutrient dynamics
- Soil fertility → population dynamics
- **Environmental heterogeneity → community dynamics**
- Invasion → mycorrhizal symbioses

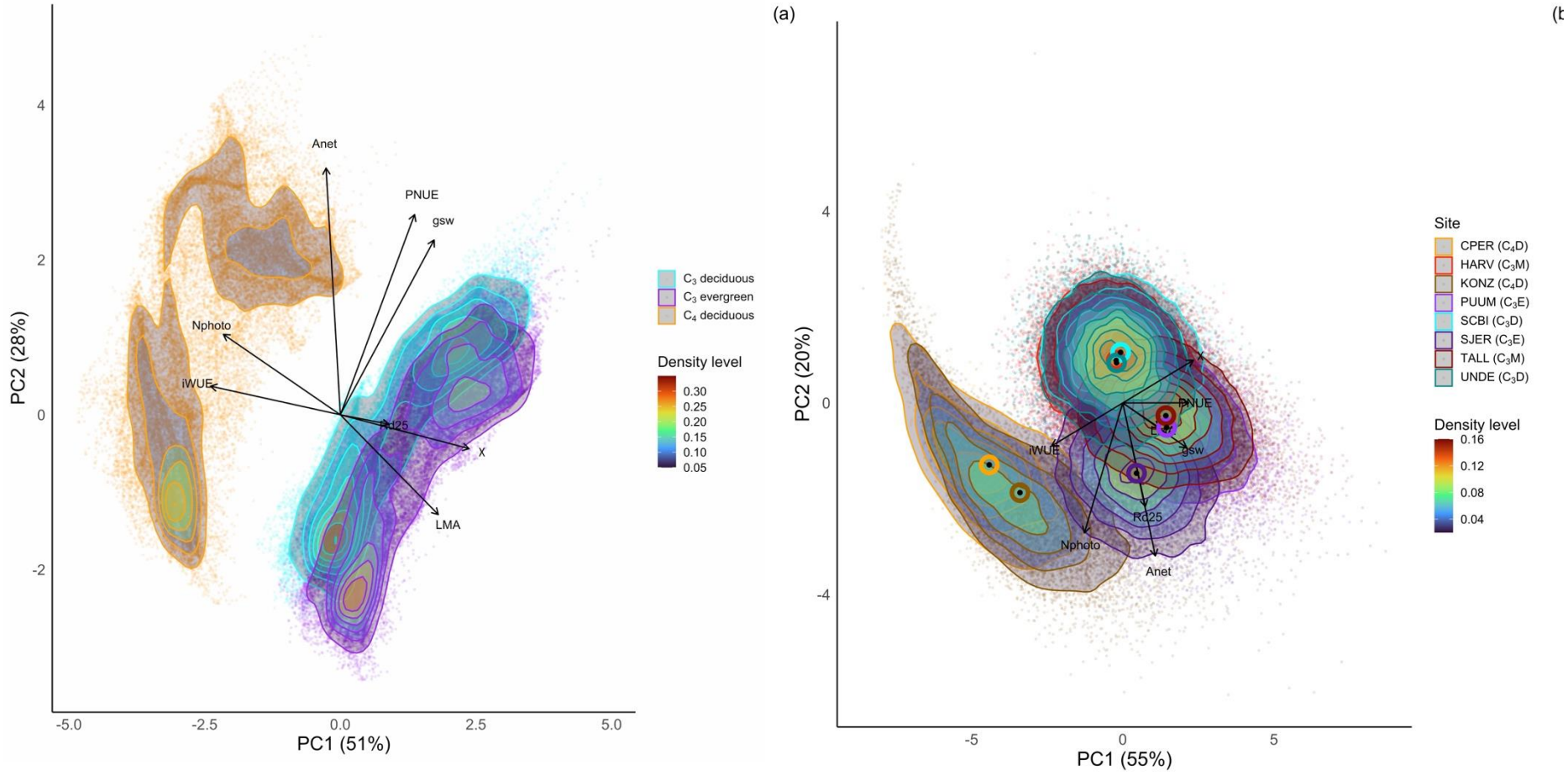

Environmental Heterogeneity
Photosynthetic Strategies

Trait strategies exist, but can we predict these?

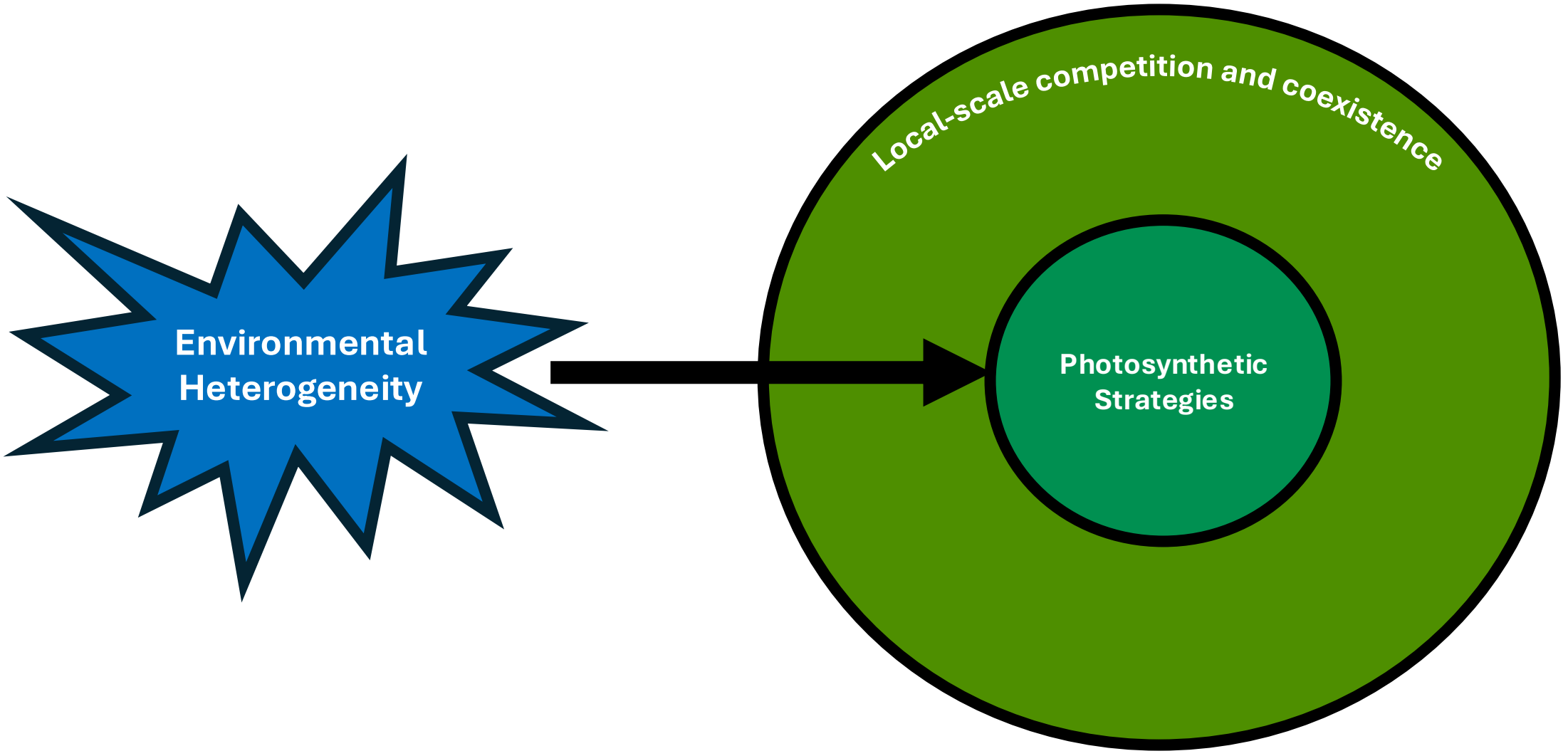


Diaz et al. (2016) *Nature*

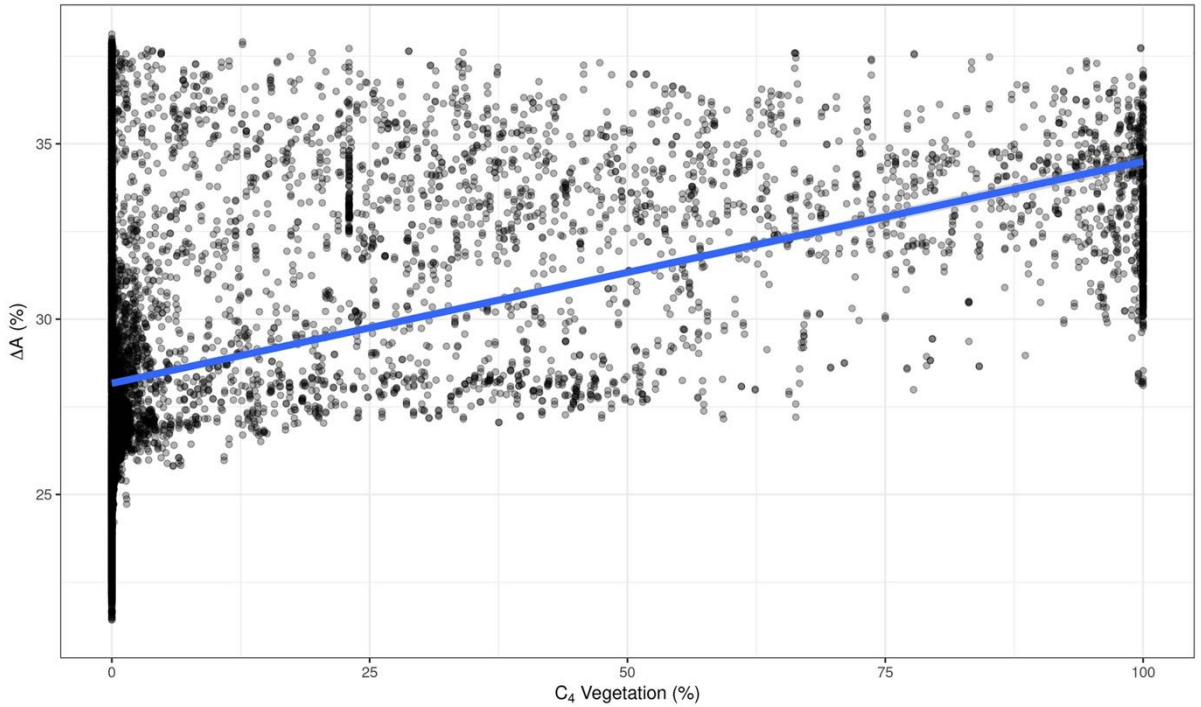
We can predict global and local photosynthetic strategies from theory

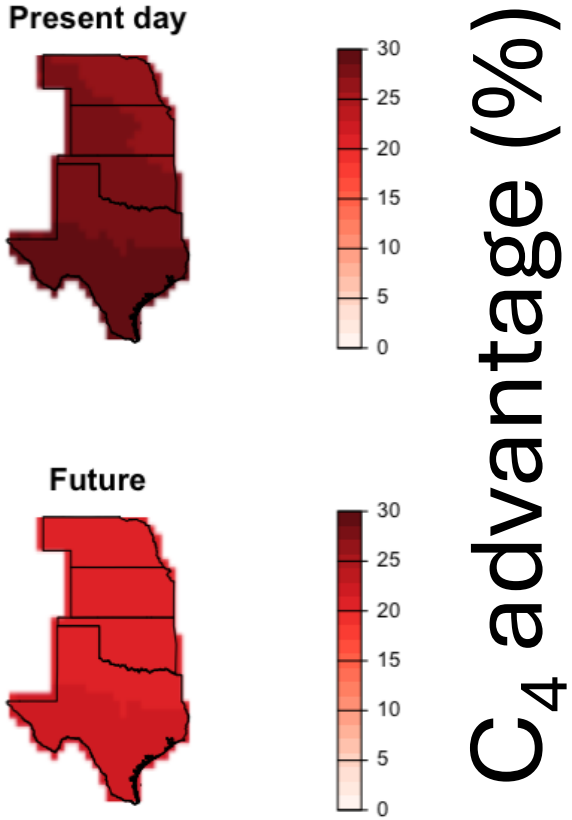


Smith (In Review)


Environmental Heterogeneity
Local-scale competition and coexistence
Photosynthetic Strategies

Strategies can predict observable outcomes of competition (broadly)







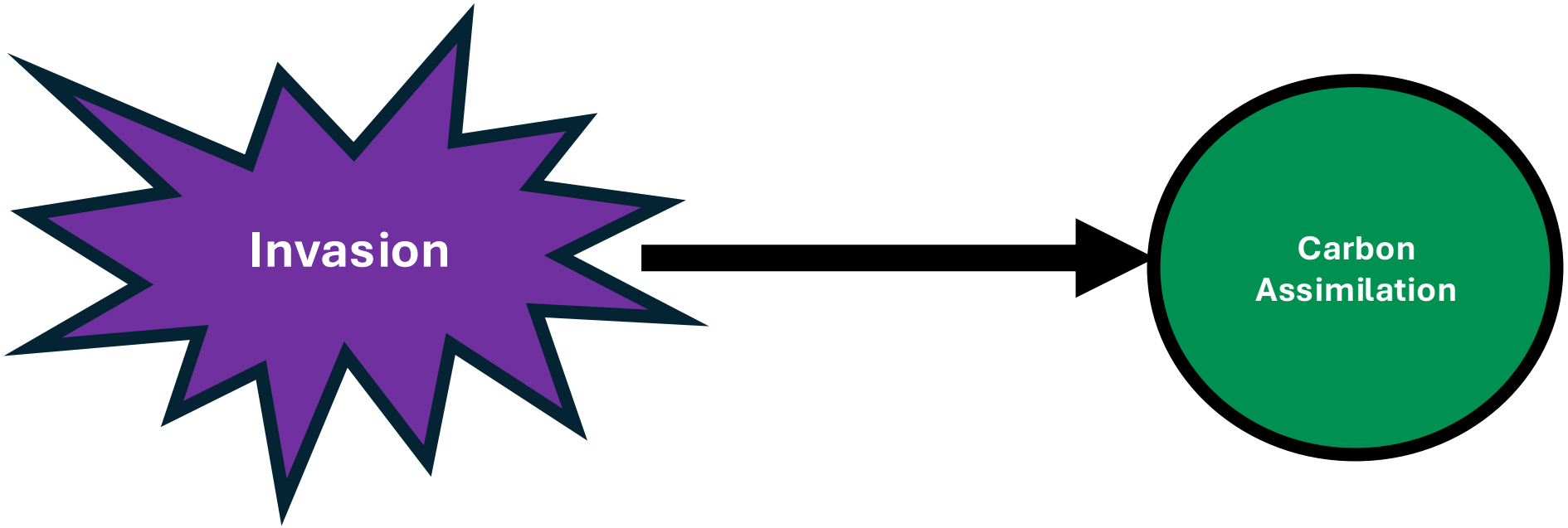




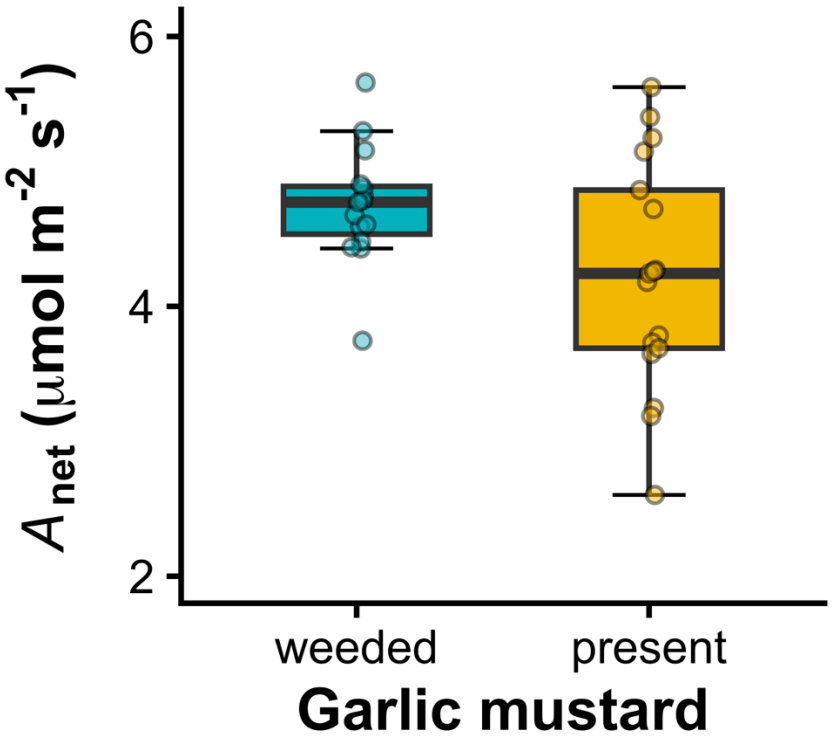
Scott and Smith (2022) *JAMES*

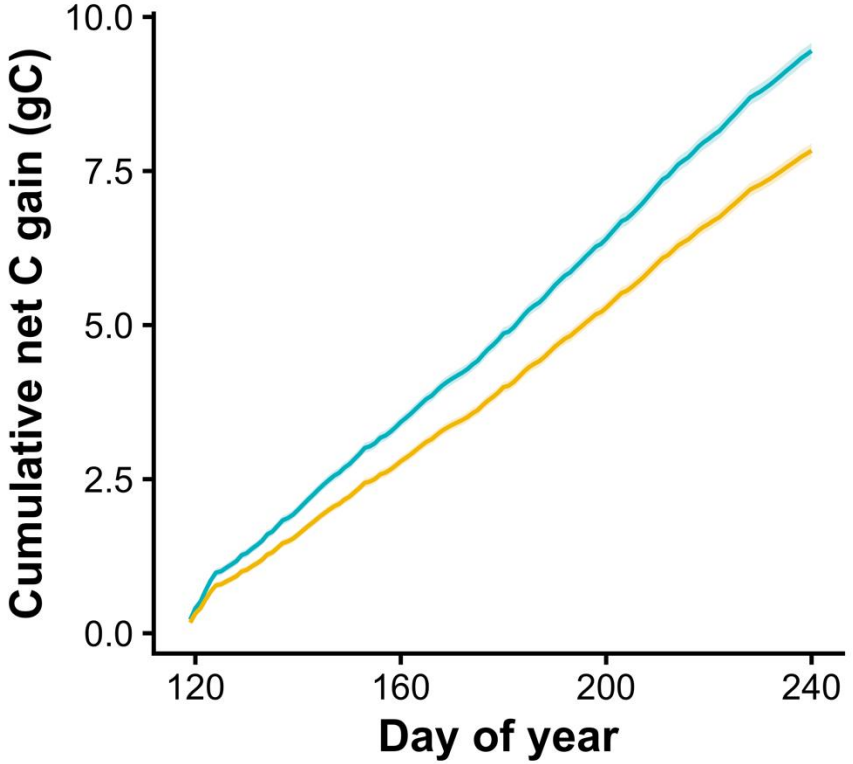
Perturbation vignettes

- Temperature → global carbon and energy fluxes
- CO_2 → plant-nutrient dynamics
- Soil fertility → population dynamics
- Environmental heterogeneity → community dynamics
- **Invasion → mycorrhizal symbioses**


Invasion
Carbon
Assimilation

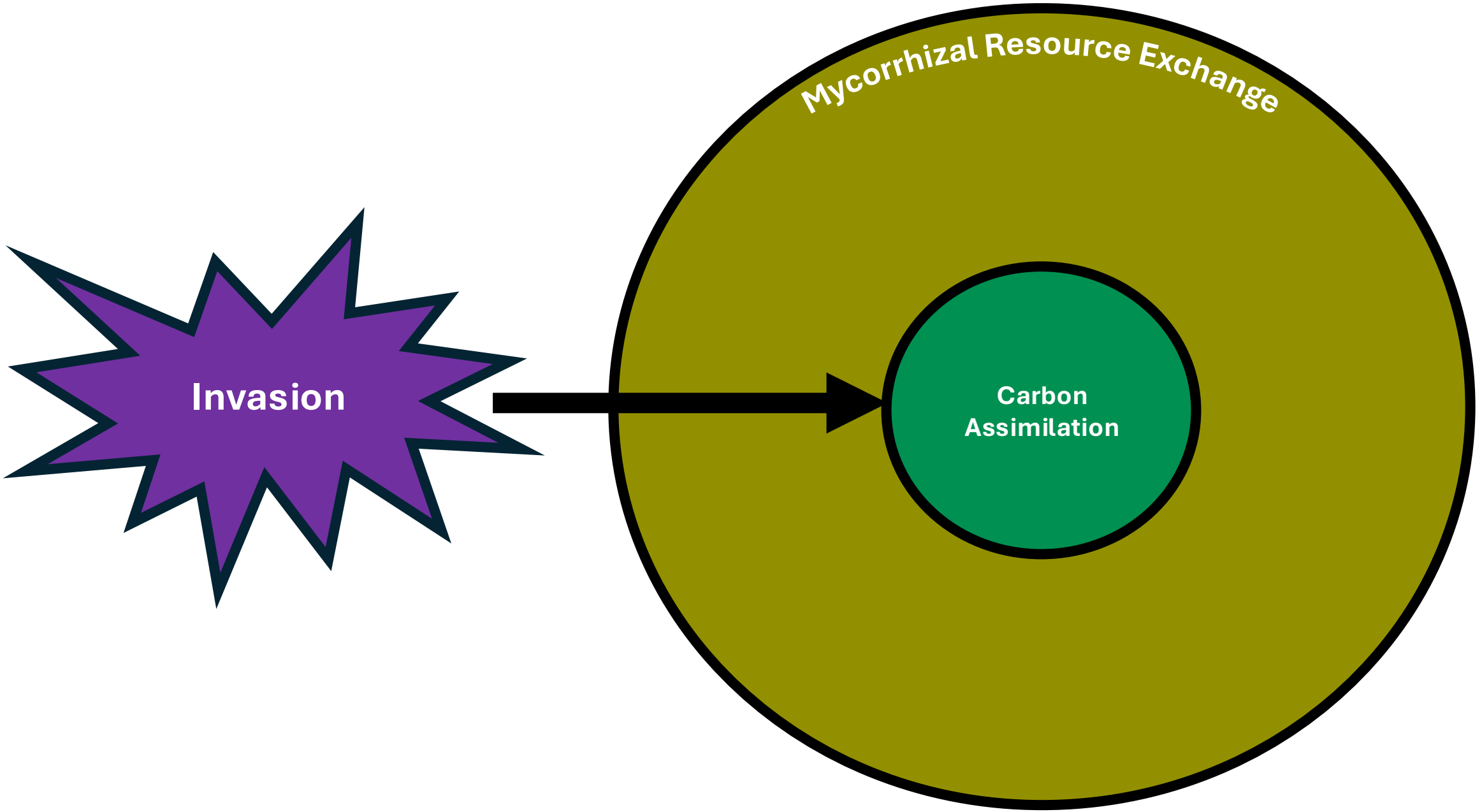
Invasion by garlic mustard reduces native understory plant carbon assimilation





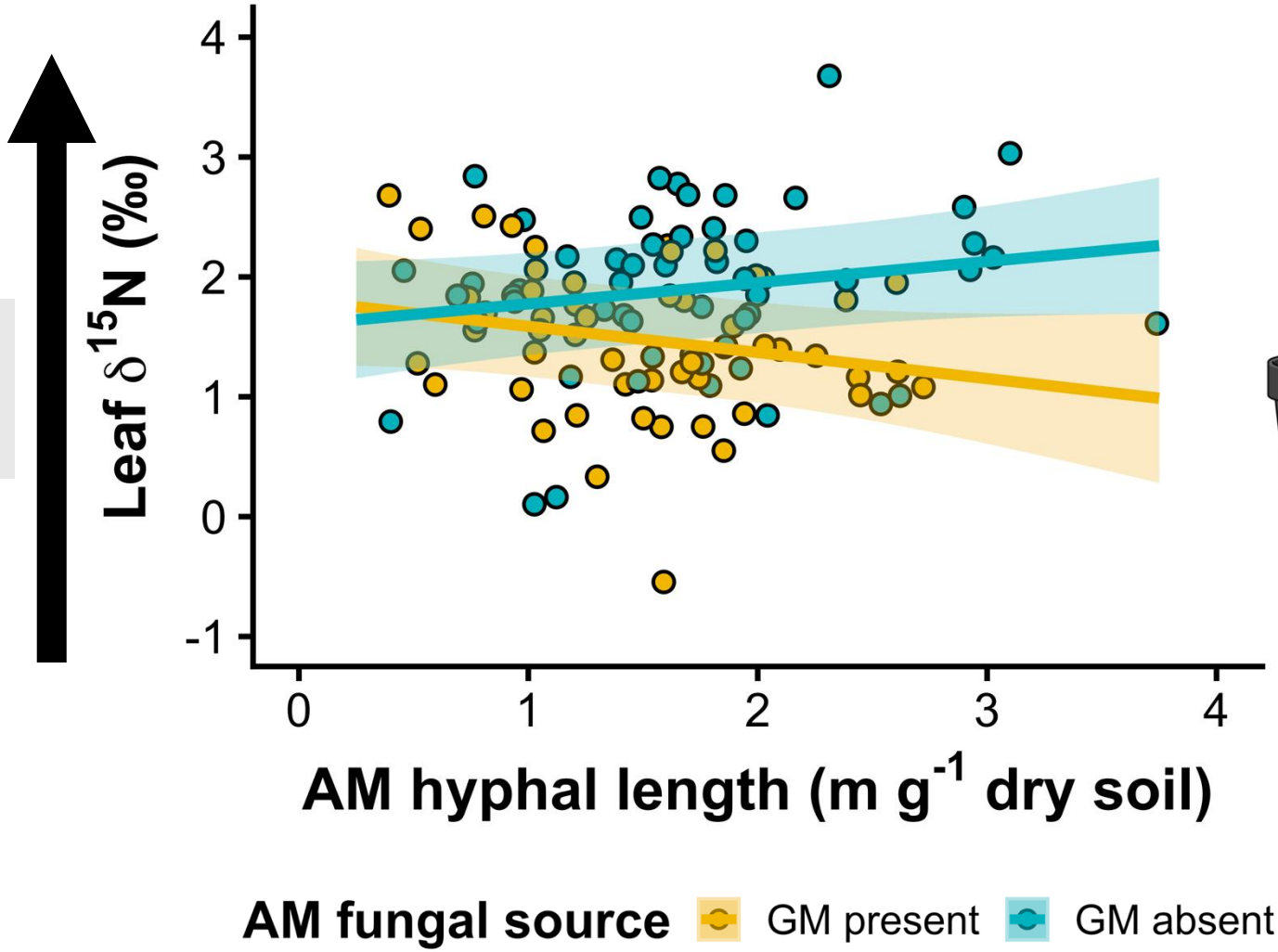


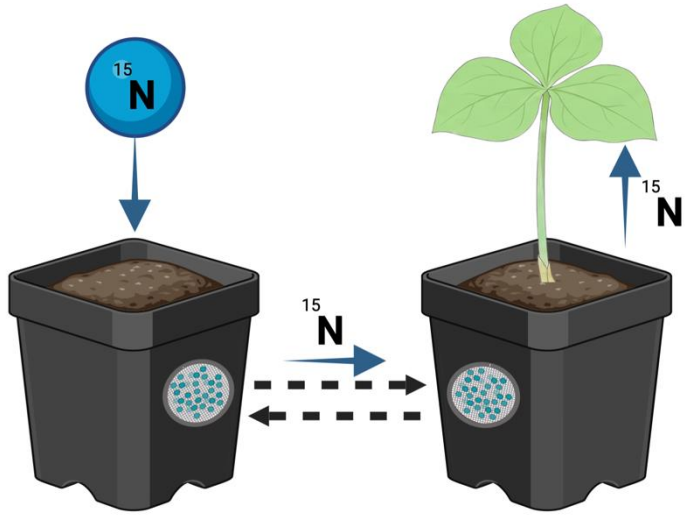
Perkowski…and Smith (2026) *Func Ecol*


Mycorrhizal Resource Exchange
Invasion
Carbon
Assimilation

Invasion impacts are due to mycorrhizal mutualism disruption

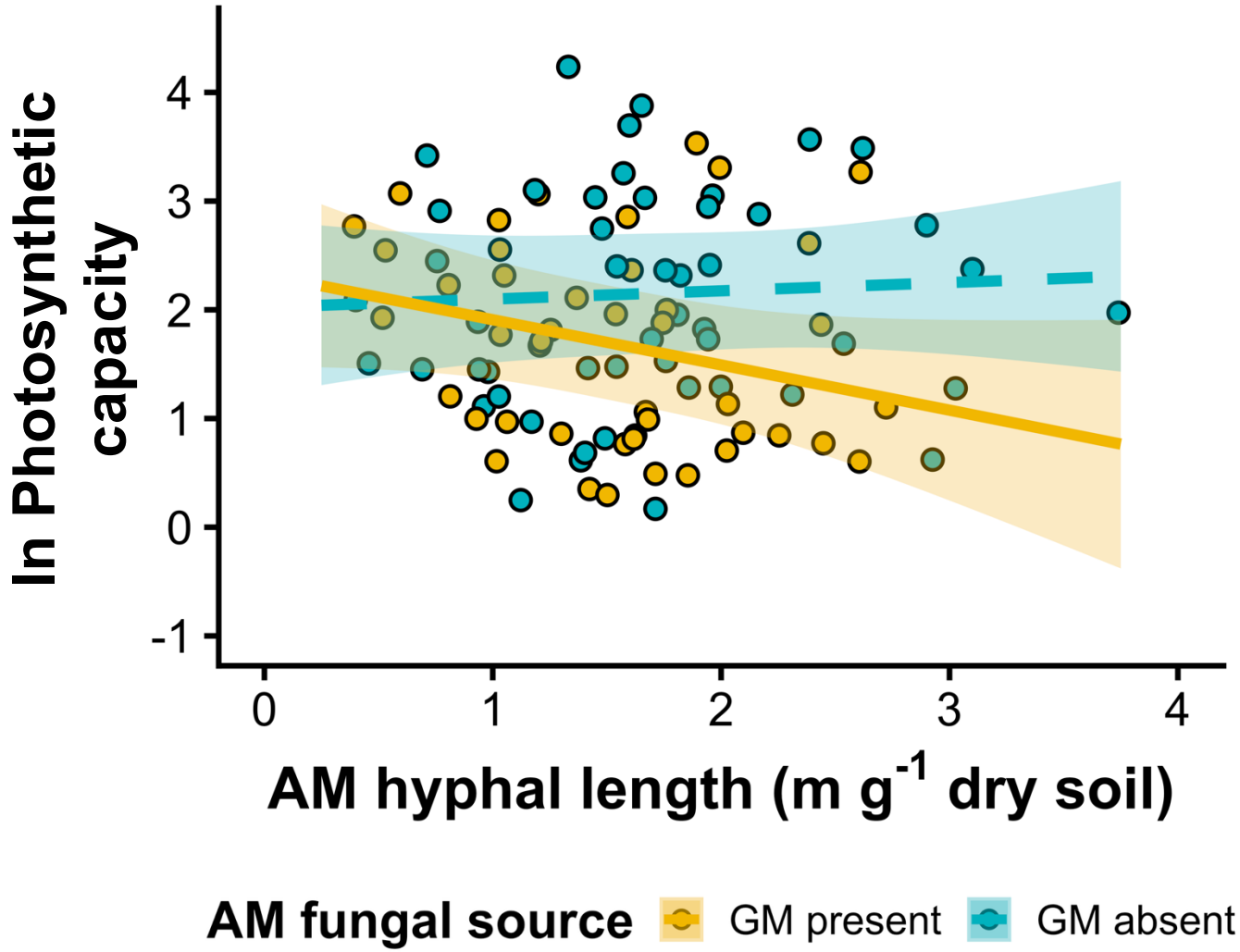
Greater share of leaf N from AM fungi





Perkowski…and Smith (In Prep)

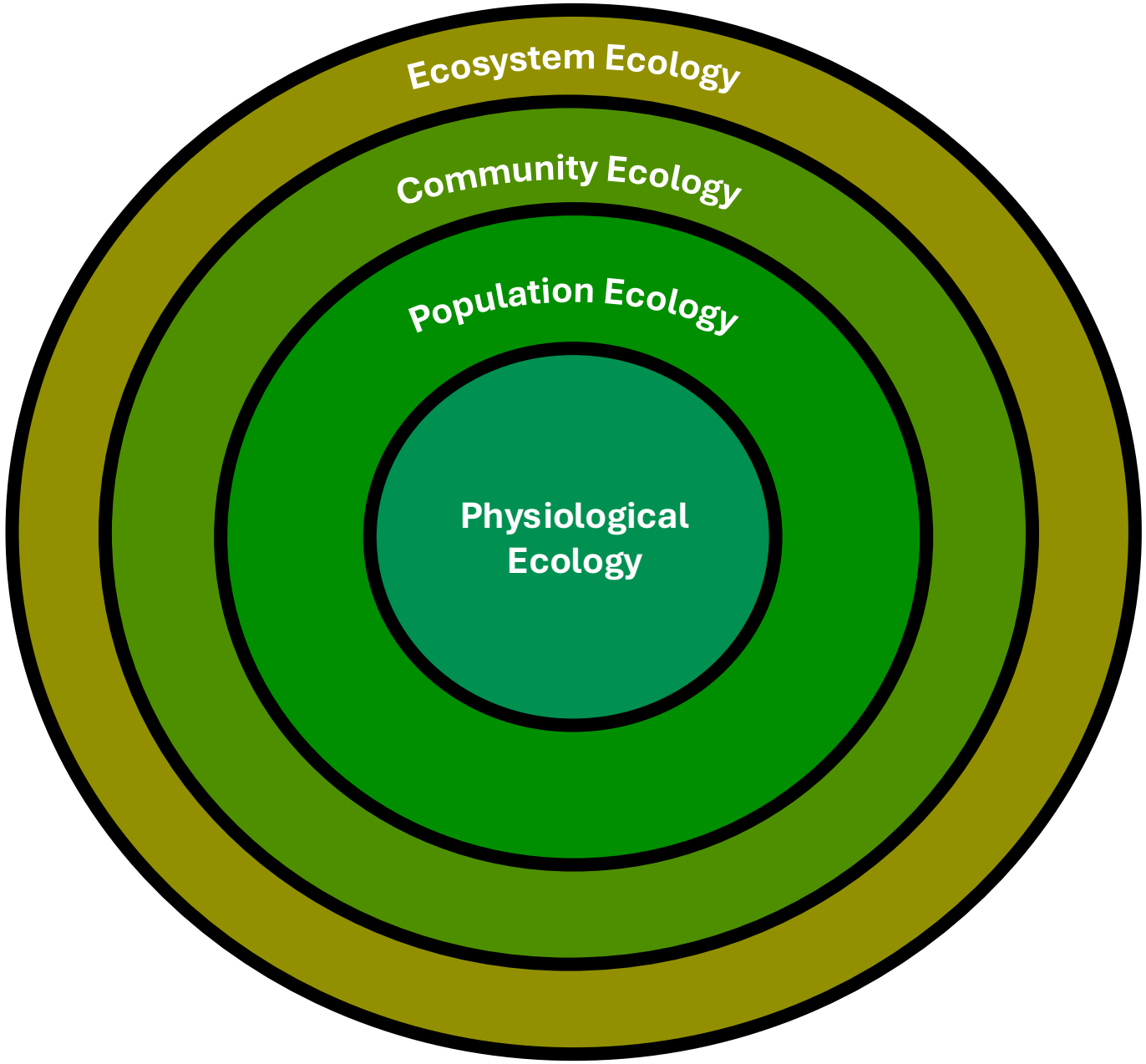
Invasion impacts are due to mycorrhizal mutualism disruption



Perkowski…and Smith (In Prep)

Conclusions

- Plant physiological responses to their environment form the basis of larger-scale responses
- Experiments, observations, theory, and modeling are needed to understand and predict these responses across scales








Presentation available at: github.com/SmithEcophysLab/seminar/
More about the lab: smithecophyslab.com
Contact: nsmith87@unl.edu

